The stator quadrature axis voltage in synchronous frames is

$$v_{qs}^e = (R_s + L_s p)i_{qs}^e + \omega_s L_s i_{ds}^e + L_m p i_{qr}^e + \omega_s L_m i_{dr}^e$$
$$= R_s i_{qs}^e + p\lambda_{qs}^e + \omega_s \lambda_{ds}^e$$

where

$$\lambda_{qs}^e = L_s i_{qs}^e + L_m i_{qr}^e$$
$$\lambda_{ds}^e = L_s i_{ds}^e + L_m i_{dr}^e$$

Substituting for the rotor q and d axes currents in terms of the stator q and d axes currents and rotor flux linkages, the d and q axes stator flux linkages are

$$\lambda_{qs}^e = \frac{L_s L_r - L_m^2}{L_r} i_{qs}^e = \sigma L_s i_{qs}^e$$

$$\lambda_{ds}^e = L_s i_{ds}^e + L_m \frac{\lambda_r - L_m i_{ds}^e}{L_r} = \frac{L_m}{L_r}\lambda_r + \frac{L_s L_r - L_m^2}{L_r} i_{ds}^e$$

where σ is the leakage coefficient of the induction machine, defined as

$$\sigma = 1 - \frac{L_m^2}{L_s L_r}$$

but in steady state, the rotor flux-linkages phasor is given by

$$\lambda_r = L_m i_{ds}^e$$

which, upon substitution, reduces the d axis stator flux linkages:

$$\lambda_{ds}^e = L_s i_{ds}^e$$

resulting in the q axis voltage:

$$v_{qs}^e = R_s i_{qs}^e + \sigma L_s p i_{qs}^e + \omega_s L_s i_{ds}^e$$

This gives the time constant in the q axis current path, i.e., i_T path, which is the torque channel in the induction machine:

$$\tau_T = \sigma \frac{L_s}{R_s} = \sigma \tau_s = \tau_s' = \text{stator-transient time constant}$$

Note that it is of the same order, a few ms, as the armature time constant in dc machines.

8.7 TUNING OF THE VECTOR CONTROLLER

The tuning of the vector controller requires the exact values of rotor resistance, mutual inductance, and rotor self-inductance of the induction machine. Implementing these values in the vector controller with proper normalization for the torque and flux commands completes the tuning task. The tuning task is simple if the motor parameters remain constant. The fact that the rotor resistance changes with temperature and frequency and the leakage inductance changes with the magnitude of the stator current complicates the tuning problem. The natural question is, then, what values are to be instrumented in the vector controller? Changes in the rotor resistance are of paramount importance. Hence, it should be possible to have the value of rotor resistance adjusted or to keep it as a nominal value (most usually a value corresponding to some steady-state operating point). The need for parameter compensation becomes apparent when the consequences of parameter changes are studied with the vector controller having fixed parameters. This is taken up in later sections.

Example 8.5

An induction motor with the following data is to be used with an indirect vector controller:

$$5 \text{ hp, Y} - \text{connected}, 3\ \phi, 60\text{ Hz}, 4\text{ poles}, 200\text{ V}. R_s = 0.277\ \Omega, R_r = 0.183\ \Omega,$$
$$L_m = 0.0538\text{ H}, L_r = 0.05606\text{ H}, L_s = 0.0553\text{ H}, J = 0.01667\text{ kg-m}^2.$$
$$\text{Rated speed} = 1766.9\text{ rpm}.$$

Find the rated rotor flux linkages and torque commands and the corresponding flux- and torque-producing components of the stator-current command, the stator-current phasor command, the torque-angle command, and the slip-speed command. The drive is assumed to be a torque amplifier.

Solution The q and d axes voltages in synchronous reference frames are

$$v_{qs}^e = V_s\frac{\sqrt{2}}{\sqrt{3}} = 163.3\text{ V and } v_{ds}^e = 0 \text{ as it is equal to the peak value of phase } a \text{ voltage.}$$

The rated speed is 1766.9 rpm; in mechanical radians, it is

$$\omega_m = \frac{2\pi N_r}{60} = 185.029\text{ rad/sec}$$

The steady-state flux linkages are evaluated from the steady-state currents; they, in turn, are found by using synchronous reference frame equations with the substitution of $p = 0$ and with the slip speed being zero. (Refer to explanations in Chapter 5.) Because the slip speed is zero, the machine does not produce electromagnetic torque; thus the stator currents are utilized to produce solely the stator and rotor flux linkages.

$$\begin{bmatrix} I_{qs}^e \\ I_{ds}^e \\ I_{qr}^e \\ I_{dr}^e \end{bmatrix} = \begin{bmatrix} R_s & \omega_s L_s & 0 & \omega_s L_m \\ -\omega_s L_s & R_s & -\omega_s L_m & 0 \\ 0 & \omega_{sl} L_m & R_r & \omega_{sl} L_r \\ -\omega_{sl} L_m & 0 & -\omega_{sl} L_r & R_r \end{bmatrix}^{-1} \begin{bmatrix} v_{qs}^e \\ v_{ds}^e \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.104 \\ 7.832 \\ 0 \\ 0 \end{bmatrix}$$

Electromagnetic torque under this condition is,

$$T_e = \frac{3}{2}\cdot\frac{P}{2}\cdot L_m(I_{qs}^e \cdot I_{dr}^e - I_{ds}^e \cdot I_{qr}^e) = 0\text{ N·m}$$

Rotor flux linkages are,

$$\lambda_{qr}^e = L_m I_{qs}^e + L_r I_{qr}^e = 0.056\text{ Wb-Turn}$$
$$\lambda_{dr}^e = L_m I_{ds}^e + L_r I_{dr}^e = 0.4213\text{ Wb-Turn}$$

The resultant rotor flux linkage is,

$$\lambda_r = \sqrt{(\lambda_{qr}^e)^2 + (\lambda_{dr}^e)^2} = 0.4214\text{ Wb-Turn}$$

It is instructive to note the difference between the stator flux-linkage magnitude and that of its rotor counterpart from the following calculations:

$$\lambda^e_{qs} = L_s I^e_{qs} + L_m I^e_{qr} = 0.0058 \text{ Wb} - \text{Turn}$$
$$\lambda^e_{ds} = L_s I^e_{ds} + L_r I^e_{dr} = 0.4331 \text{ Wb} - \text{Turn}$$

and the stator flux linkage is

$$\lambda_s = \sqrt{(\lambda^e_{qs})^2 + (\lambda^e_{ds})^2} = 0.4331 \text{ Wb} - \text{Turn}$$

which is approximately 2.8% higher than the rotor flux linkages

The stator-current phasor magnitude is

$$I_s = \sqrt{(I^e_{ds})^2 + (I^e_{qs})^2} = \sqrt{(0.104)^2 + (7.832)^2} = 7.832 \text{ A} = I_f$$

which is equal to the flux-producing stator current in the machine and is denoted as I_f. Note that this is peak value and not rms value. The friction and windage losses are not given, so they can be neglected. Therefore, the electromagnetic torque is equal to shaft torque; its rated value is obtained as

$$T_e = \frac{P_o}{\omega_m} = \frac{5*745.6}{185.029} = 20.178 \text{ N·m}$$

The torque constant K_{te} is

$$K_{te} = \frac{3}{2} \cdot \frac{P}{2} \cdot \frac{L_m}{L_r} = 2.88$$

and, by using this, the torque-producing component of the stator current and the stator-current phasor are obtained as

$$I_t = \frac{T_e}{K_{te}\lambda_r} = \frac{20.178}{2.882 \times 0.4213} = 16.63 \text{ A}$$
$$I_s = \sqrt{I_f^2 + I_t^2} = 18.38 \text{ A}$$

The torque angle is

$$\theta_T = \tan^{-1}\left\{\frac{I_t}{I_f}\right\} = 1.13 \text{ rad}$$

The slip speed is verified from the above as

$$\omega_{sl} = \frac{R_r}{L_r} \cdot \frac{I_t}{I_f} = \frac{0.183}{0.056} \cdot \frac{16.63}{7.832} = 6.932 \text{ rad/sec}$$

The actual variables will equal their command values in steady state. Therefore, the computed variable values are the respective command values for rated operation.

Example 8.6

Consider Example 8.5. Compute the stator-voltage magnitude at rated operating conditions with rated rotor flux linkages, and comment on the finding.

Solution The steady-state conditions, determined in Example 8.5 to produce rated torque at rated speed and rated rotor flux linkages, are

$$I_T = I_{qs}^e = 16.63 \text{ A}$$
$$I_f = I_{ds}^e = 7.832 \text{ A}$$
$$\omega_{sl} = 6.932 \text{ rad/sec}$$

From the steady-state rotor equations, the rotor currents are found as follows:

$$\begin{bmatrix} I_{qr}^e \\ I_{dr}^e \end{bmatrix} = \omega_{sl} L_m \begin{bmatrix} R_r & \omega_{sl} L_r \\ -\omega_{sl} L_r & R_r \end{bmatrix}^{-1} \begin{bmatrix} -I_{ds}^e \\ I_{qs}^e \end{bmatrix}$$
$$I_{qr}^e = -15.96 \text{ A}$$
$$I_{dr}^e = 0$$

Then the stator voltages are computed from the stator steady-state equations as

$$V_{qs}^e = R_s I_{qs}^e + \omega_s L_s I_{ds}^e + \omega_s L_m I_{dr}^e = 167.85 \text{ V}$$
$$V_{ds}^e = R_s I_{ds}^e - \omega_s L_s I_{qs}^e - \omega_s L_m I_{qr}^e = -20.84 \text{ V}$$

The magnitude of stator voltage is

$$V_s = \sqrt{(V_{qs}^e)^2 + (V_{ds}^e)^2} = 169.18 \text{ V}$$

which, in terms of the line-to-line voltage, is given by

$$V = \frac{\sqrt{3}}{\sqrt{2}} V_s = 207.2 \text{ V}$$

It is 7.2 V higher than the rated value. Therefore, to operate within rated voltage and to deliver rated power, the rotor flux linkage has to be decreased. Flux-weakening is treated in a later section.

8.8 FLOWCHART FOR DYNAMICS COMPUTATION

It is useful to determine the transient response of the indirect vector-controlled induction motor, in order to evaluate its application to a specific requirement. Also, such a study is a vehicle for avoiding the costly process of prototype construction and testing. The guidelines adopted for the flowchart are that the machine and vector controller parameters are specified along with the inverter and switching logic. Note that any converter can be used in the simulation. The system equations are assembled by starting from the speed controller, vector controller, inverter, switching logic, and the machine with the currents, speed, and position feedbacks. The speed command and torque disturbance need to be specified to evaluate the response. The nonlinear equations of the machine are used in the simulation, and stator dynamics are included. The flowchart is given in Figure 8.25.

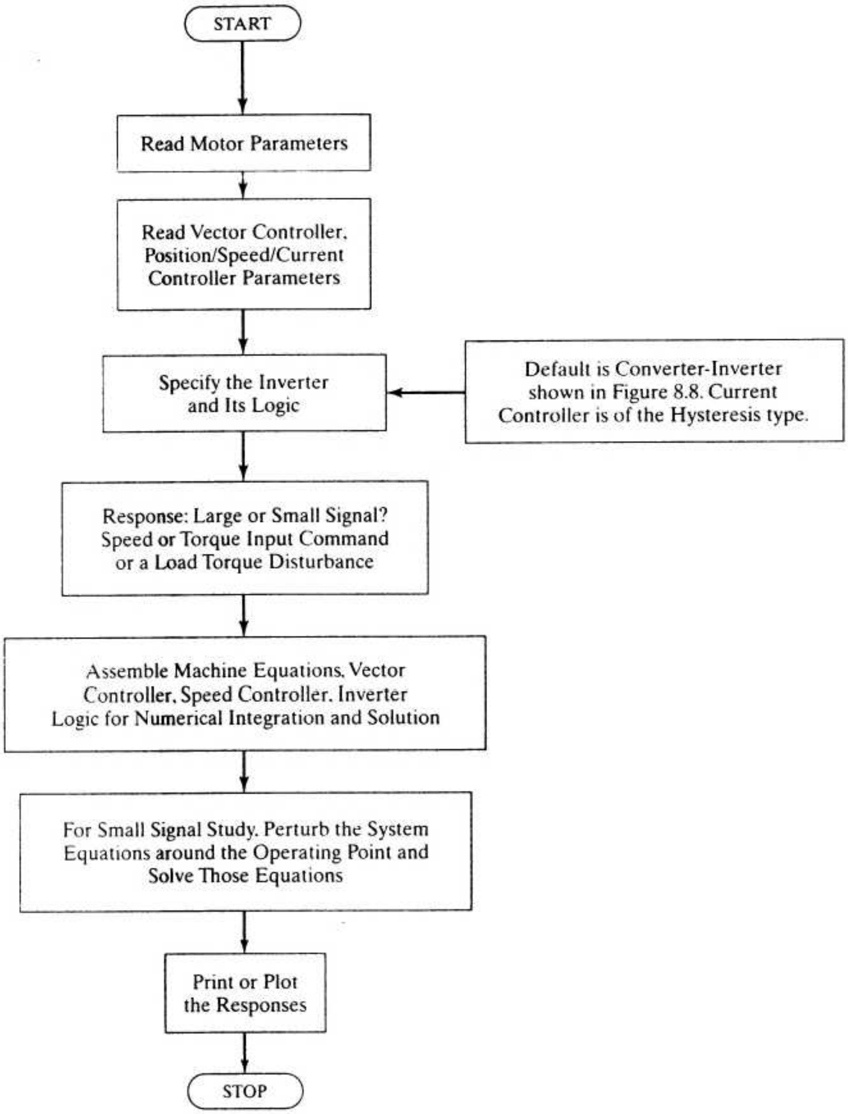


Figure 8.25 Flowchart for dynamic simulation of the vector-controlled induction motor drive

8.9 DYNAMIC SIMULATION RESULTS

A functional block diagram of an indirect vector-controlled induction motor drive with *dq* axes current control is shown in Figure 8.26. The phase-current feedback loops are substituted with the *qd* axes current-feedback loops, thus enabling the individual design of the respective current controllers, considering their different time constants dominating the torque and flux responses. This is the distinct advantage of such a configuration.

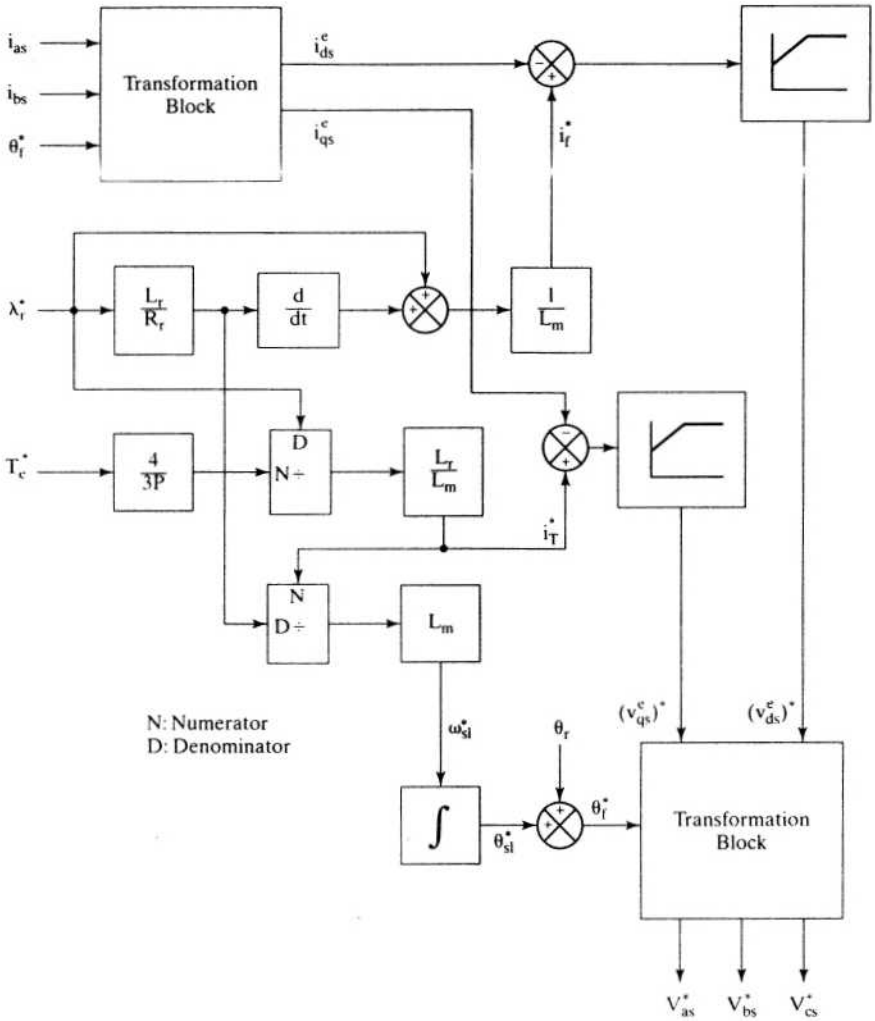


Figure 8.26 Functional block diagram of a voltage-source indirect vector controller

A 0.25-p.u. bidirectional-step speed reference is given; its response is shown in Figure 8.27. The electromagnetic torque command is limited to 2 p.u., and rotor flux linkage is at rated value at the time of application of the speed command. The response of the torque is almost instantaneous, and the speed response is very smooth without ripples. Further, the drive goes through the first-, fourth-, and third-quadrant operation without any current, flux, and torque oscillations. The drive uses PWM voltage control at the final stage of the controller.

The performance of the drive system for a sinusoidal speed reference is shown in Figure 8.28. Small-signal sinusoidal response is helpful to determine the speed loop bandwidth of the system. For a low-frequency sinusoidal command, note that the response is a true replica of the reference itself. For higher frequencies of the reference, both attenuation and phase shift occur in the responses.

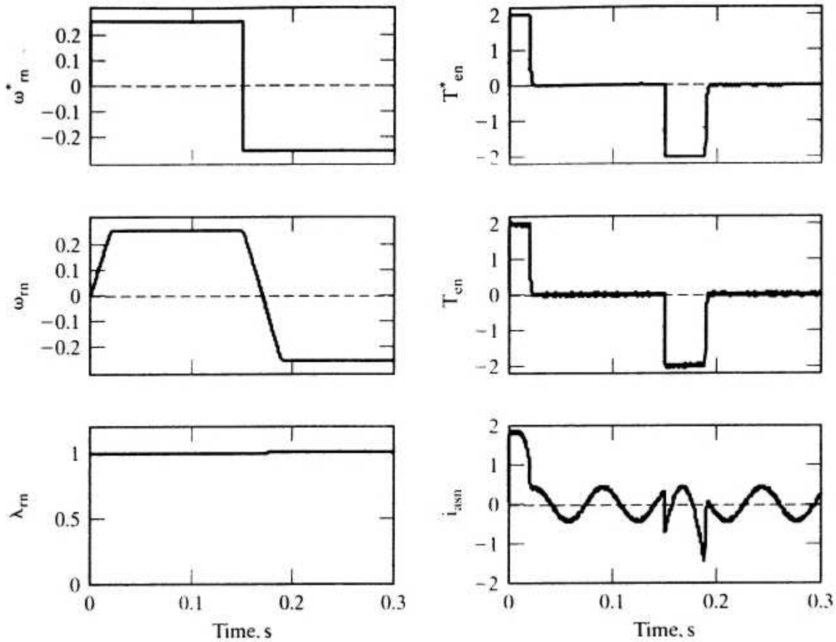


Figure 8.27 Output characteristics for a speed-controlled voltage-source indirect vector-control scheme in p.u.

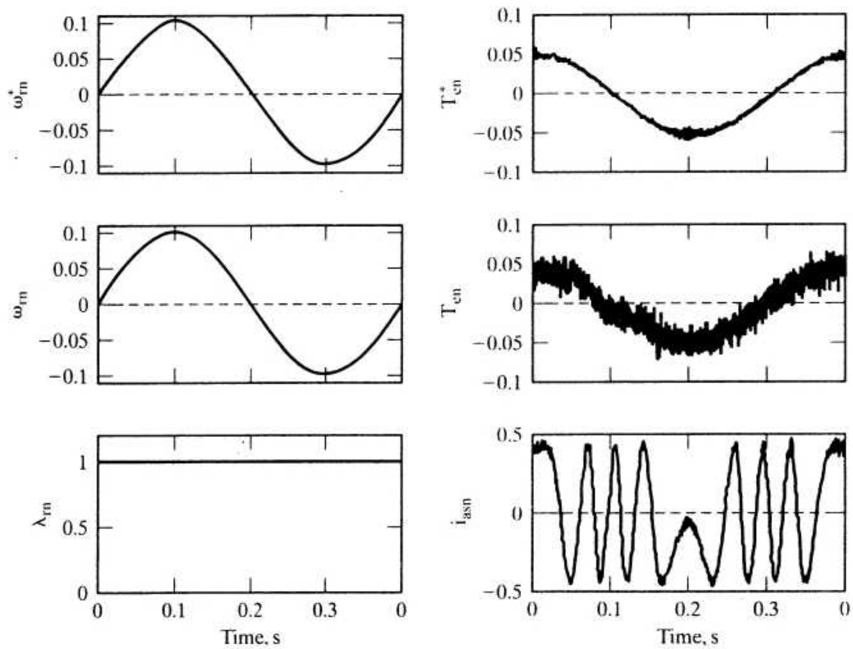


Figure 8.28 Sinusoidal speed response of a current-source indirect vector-controlled induction motor drive in p.u.

8.10 PARAMETER SENSITIVITY OF THE INDIRECT VECTOR-CONTROLLED INDUCTION MOTOR DRIVE

A mismatch between the vector controller and induction motor occurs as a result either of the motor parameters changing with operating conditions such as temperature rise and saturation or of the wrong instrumentation of the parameters in the vector controller. The latter phenomenon is controllable, but the former is dependent on the operating conditions of the motor drive and hence is uncontrollable. The mismatch produces a coupling between the flux- and torque-producing channels in the machine. This has the following consequences:

(i) The rotor flux linkages deviate from the commanded value.

(ii) The electromagnetic torque, hence, deviates from its commanded value, producing a nonlinear relationship between the actual torque and its commanded value.

(iii) During torque transients, an oscillation is caused both in the rotor flux linkages and in torque responses, with a settling time equal to the rotor time constant. The rotor time constant is large: on the order of 0.5 second or greater.

In a torque drive, consequences (ii) and (iii) are most undesirable. Although, in the speed-controlled drive, the nonlinear torque-to-torque command characteristic will not have a detrimental effect on the steady-state operation, its effect is considerable during the transients. The load and motor inertia are required to smooth these torque excursions so that they do not appear as speed ripples. For the present, the parameter sensitivity effects are quantified in the steady state and the transient state, considering both open and closed outer speed loop.

8.10.1 Parameter Sensitivity Effects When the Outer Speed Loop Is Open

With the outer speed loop open in the vector-controlled induction motor drive, the external commands are the rotor flux linkages and electromagnetic torque. The actual slip speed and commanded value are equal in this mode of the drive. Therefore, the mismatch between the vector controller and induction motor induces deviations in the flux- and torque-producing stator-current components and hence in the torque angle. The net effects are the deviations in the magnitude of the rotor flux linkages and electromagnetic torque. They are derived in the following sections.

8.10.1.1 Expression for electromagnetic torque. In the steady state, an expression for the rotor flux linkages is obtained by substituting $p = 0$ in equation (8.56):

$$\lambda_r^* = L_m^* i_f^* \tag{8.64}$$

Substituting equation (8.64) into equation (8.57) gives the command value of the slip speed as

$$\omega_{sl}^* = \left[\frac{1}{T_r^*}\right]\left[\frac{i_T^*}{i_f^*}\right] \tag{8.65}$$

The torque-angle command is known to be

$$\theta_T^* = \tan^{-1}\left[\frac{i_T^*}{i_f^*}\right] \tag{8.66}$$

Substituting equation (8.66) into (8.65) cause the torque-angle-to-slip-speed relationship to emerge as

$$\tan \theta_T^* = \omega_{sl}^* T_r^* \tag{8.67}$$

from which the sine and cosine of the torque angle are found as

$$\sin \theta_T^* = \frac{\omega_{sl}^* T_r^*}{\sqrt{1 + (\omega_{sl}^* T_r^*)^2}} \tag{8.68}$$

$$\cos \theta_T^* = \frac{1}{\sqrt{1 + (\omega_{sl}^* T_r^*)^2}} \tag{8.69}$$

The command value of the torque, from related equations, could be written as

$$T_e^* = K_{te}\lambda_r^* i_T^* = K_{te} L_m^* i_T^* i_f^* = \frac{1}{K_{it}} \frac{(L_m^*)^2}{L_r^*} (i_s^*)^2 \cos \theta_T^* \sin \theta_T^* \tag{8.70}$$

Likewise, the actual electromagnetic torque generated in the motor is expressed as

$$T_e = \frac{1}{K_{it}} \frac{L_m^2}{L_r} i_s^2 \cos \theta_T^* \sin \theta_T^* \tag{8.71}$$

because the command and actual torque angles are equal in steady state. In the torque mode, the constraints in force are

$$i_s = i_s^* \tag{8.72}$$

$$\omega_{sl} = \omega_{sl}^* \tag{8.73}$$

Substituting equations (8.67), (8.68), (8.72), and (8.73) into (8.70) and (8.71), the ratio of torque to its command value is obtained:

$$\frac{T_e}{T_e^*} = \left[\frac{L_m}{L_m^*}\right]^2 \left[\frac{L_r^*}{L_r}\right] \left[\frac{T_r}{T_r^*}\right] \left[\frac{1 + (\omega_{sl}^* T_r^*)^2}{1 + (\omega_{sl}^* T_r)^2}\right] \tag{8.74}$$

Defining the nondimensional variables between the motor (actual) and controller-instrumented rotor time constants and magnetizing inductances to generalize the results and their interpretation, we get

$$\alpha = \frac{T_r}{T_r^*} \tag{8.75}$$

$$\beta = \frac{L_m}{L_m^*} \tag{8.76}$$

and making the approximation that the leakage inductance is negligible compared to the mutual inductance leads to the elimination of one more parameters, resulting in a simpler and compact representation of the parameter sensitivity effects:

$$\frac{L_r^*}{L_r} \cong \frac{L_m^*}{L_m} = \frac{1}{\beta} \tag{8.77}$$

Substituting equations from (8.75) to (8.77) into equation (8.74) gives the actual-to-command torque value in terms of the normalized values α and β as

$$\frac{T_e}{T_e^*} = \alpha\beta\left[\frac{1 + (\omega_{sl}^* T_r^*)^2}{1 + (\alpha\omega_{sl}^* T_r^*)^2}\right] \tag{8.78}$$

Note that the factors α and β embody the aspects of controller mismatch, saturation, and machine temperature, as reflected in the rotor resistance.

8.10.1.2 Expression for the rotor flux linkages. The parameter variations affect primarily the rotor flux linkages. The change in steady-state rotor flux linkages gives an indication of the change in the electromagnetic torque. In steady state, the actual and the commanded rotor flux linkages are

$$\lambda_r^* = L_m^* i_f^* \tag{8.79}$$

$$\lambda_r = L_m i_f \tag{8.80}$$

Substituting for i_f in terms of stator current and finding the ratio of the actual to command rotor flux linkages gives

$$\frac{\lambda_r}{\lambda_r^*} = \left[\frac{L_m}{L_m^*}\right]\left[\frac{\cos\theta_T}{\cos\theta_T^*}\right] = \left[\frac{L_m}{L_m^*}\right]\sqrt{\frac{1 + (\omega_{sl}^* T_r^*)^2}{1 + (\alpha\omega_{sl}^* T_r^*)^2}} = \beta\sqrt{\frac{1 + (\omega_{sl}^* T_r^*)^2}{1 + (\alpha\omega_{sl}^* T_r^*)^2}} \tag{8.81}$$

A set of normalized curves between the torque ratio, flux ratio, and α for various slip speeds and saturation levels β are presented in the following section.

8.10.1.3 Steady-state results. The rationale for choosing the range of values for α and β is discussed before the results are presented.

Ranges of α and β The practical temperature excursion of the rotor is approximately 130°C above ambient. This increases the rotor resistance by 50% over its ambient or nominal value. Magnetic saturation can decrease the self-inductance to 80% of its nominal value. Hence, the lower limit of α can be obtained from the following:

$$T_r = \frac{0.8L_m^*}{1.5R_r^*} \cong 0.533T_r^* \tag{8.82}$$

Since $\alpha = T_r/T_r^*$, the lowest value of α is approximately 0.5. The upper limit of α approaches 1.5, depending primarily on errors in the instrumented vector controller and an increase in rotor self-inductance because of unsaturated operating point on the BH material characteristics of the laminations during flux weakening. Hence, the range of value for α is chosen to be

$$0.5 < \alpha < 1.5 \tag{8.83}$$

A typical value of β in the magnetic saturation region is around 0.8, and operation of the induction motor drive in the linear portion of the iron B-H characteristics

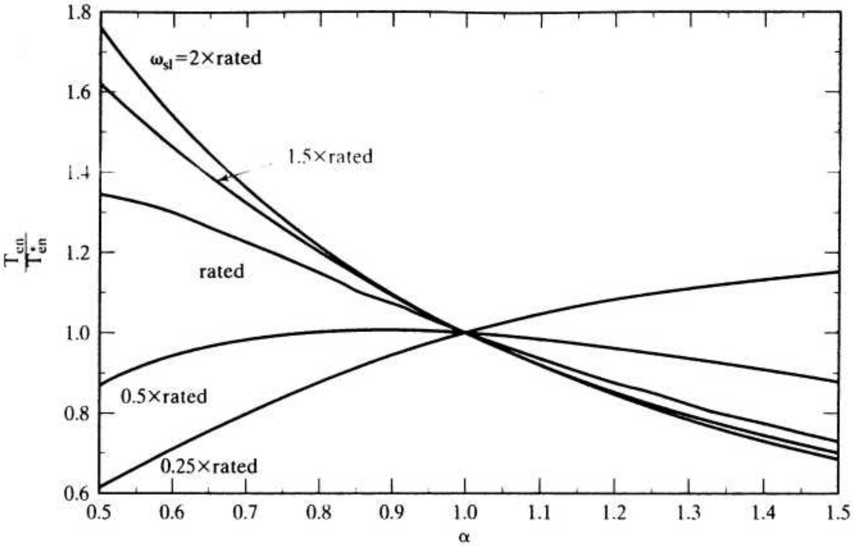


Figure 8.29(i) Torque and its command versus α for various slip speeds

increases β to 1.2. Note that these values are given for stator currents not exceeding twice the rated value. Hence, the range of values for β is chosen as

$$0.8 < \beta < 1.2 \tag{8.84}$$

Steady-State Torque and Flux The steady-state responses for commands of torque and rotor flux linkages as a function of α are shown in Figures 8.29(i) and 8.29(ii), respectively. In these figures, the saturation factor β is maintained at 1 while the slip speed is varied from rated to three times rated value. For $\alpha < 1$, mainly signifying the increase in rotor temperature, the rotor flux linkages and the electromagnetic torque are greater than their command values. Though the rotor flux is shown to be increasing to value exceeding 1.2 p.u., it is difficult to maintain this high value, because of saturation. Increasing motor saturation at ambient temperature increases α; for such operating points, the torque and rotor flux are less than their commanded values. Note that α increases when the induction motor drive is started at a temperature less than the assumed ambient temperature that usually is used to set the nominal vector-controller values. The same figures also delineate the characteristics resulting from a mismatch between the instrumented-controller parameters and the actual value of the motor parameters.

The linearity of the input–output torque relationship has degraded, and this induction motor drive is unsuitable for use in torque-control applications. A parameter compensator is essential to achieve a linear torque amplifier. High-performance applications, such as robotic drives, require precise control of torque; the torque takes precedence over the speed loop and comes directly after the position loop. Those applications can elude the uncompensated induction motor drive. Saturation affects

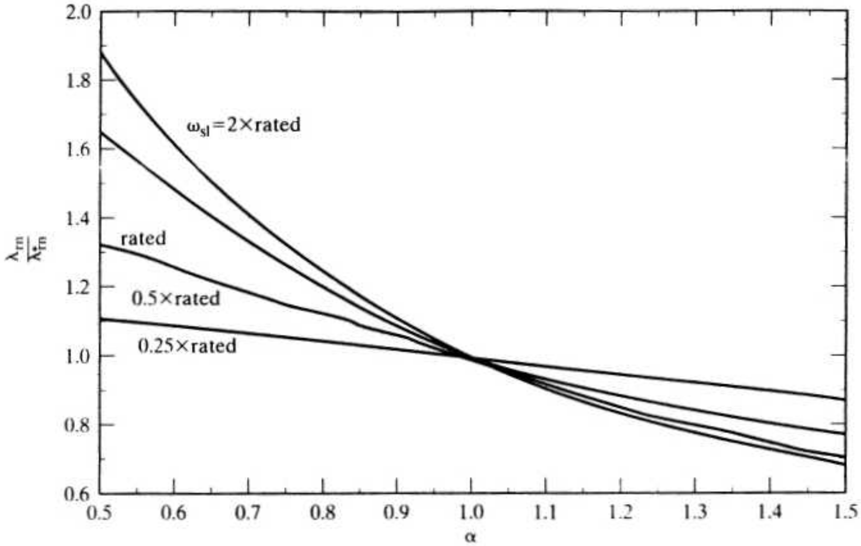


Figure 8.29(ii) Rotor flux linkage and its command versus α for various slip speeds

the torque and flux proportionally, as is seen from equations (8.78) and (8.81); hence, it will not be elaborated any further.

In general, increasing temperature causes saturation and an increase in the output torque. However, for torque commands less than 0.5 p.u., output torque is reduced.

Example 8.7

The induction motor given in Example 8.4 was run in the torque mode. The ratio of rotor flux linkage to its command value was observed to be 1.1. The pertinent data available for the operating point are

$$\alpha = 0.6$$
$$\beta = 0.9$$

Find the ratio of actual torque to its command value, the slip-speed command, and the torque command, if $\lambda_r^* = 0.4$ Wb-turns.

Solution

$$\frac{\lambda_r}{\lambda_r^*} = 1.1 = \beta\sqrt{\frac{1 + (\omega_{sl}^* T_r^*)^2}{1 + (\alpha\omega_{sl}^* T_r^*)^2}} = 0.9\sqrt{\frac{1 + (0.3063\omega_{sl}^*)^2}{1 + (0.6 \times 0.3063\omega_{sl}^*)^2}}$$

Rearrange this and solve for ω_{sl}^*; $\omega_{sl}^* = 3.375$ rad/s and

$$\frac{T_e}{T_e^*} = \alpha\beta\left[\frac{1 + (\omega_{sl}^* T_r^*)^2}{1 + (\alpha\omega_{sl}^* T_r^*)^2}\right] = 0.6 \times 0.9\left[\frac{1 + (0.3063 \times 3.375)^2}{1 + (0.6 \times 0.3063 \times 3.375)^2}\right] = 0.806$$

$$\omega_{sl}^* = \frac{L_m^* i_T^*}{T_r^* \lambda_r^*}$$

from which the torque current and torque command are obtained as

$$i_T^* = \frac{\omega_{sl}^* \lambda_r^* T_r^*}{L_m^*} = \frac{3.375 \times 0.4 \times 0.3063}{0.0538} = 7.696\ \text{A}$$

$$T_e^* = K_t \lambda_r^* i_T^* = \frac{3}{2}\frac{P}{2}\frac{L_m^*}{L_r^*}\lambda_r^* i_T^* = \frac{3}{2} \times \frac{4}{2} \times \frac{0.0538}{0.05606} \times 0.4 \times 7.696 = 8.862\ \text{N·m}$$

8.10.1.4 Transient characteristics. Simulation results for the dynamic operation of the parameter-sensitive induction motor drive are obtained by solving the rotor equations given in (8.27) and (8.28) and the electromechanical equation given by

$$J\frac{d\omega_r}{dt} = \frac{P}{2}(T_e - T) \tag{8.85}$$

where J is the moment of inertia and T is the load torque.

The effect of switching delays in the control logic of the inverter and in the inverter itself is considered to be negligible. The stator dynamics are omitted, as they can be for a high-performance transistorized inverter with a 5 kHz current loop bandwidth. Experimental results have confirmed the accuracy of the above assumptions [31].

It can be shown that the flux and torque have a time constant equal to the rotor time constant when the rotor flux linkage is allowed to vary and the natural frequency of oscillation of the system corresponds with the slip speed. The system damping factors are determined by many factors, including α and β. Different behavior has been observed for the motoring and the regenerating operation of the drive on the open-speed loop. A sample simulation result is shown in Figure 8.30 for a stator temperature rise of 100°C. This figure contains the torque command, actual torque, and the rotor flux linkages for a shaft speed of 300 rpm. Note that the rotor flux linkages have not been modified for saturation. It is symbolic of the behavior for unsaturated conditions in the motor drive.

Although the developed flux and torque increase for rising temperature, the oscillatory responses are undesirable and make the induction motor drive a nonideal torque amplifier. The transient performance of the indirect vector-controlled induction motor drive is derived analytically in the following for both the rotor flux linkages and the torque responses. These derivations demonstrate the validity of the statements in the preceding discussion and help us to understand the parameters influencing the transient responses. A time delay, τ_c, involved in the currents by stator dynamics, is also incorporated, along with the parameter mismatch between the controller and motor.

Rotor Flux Linkages Response Considering parameter sensitivity in the indirect vector-controlled induction motor drive and a time lag of τ_c between the currents and their commands, the transfer function between the rotor flux linkage and its command is derived. From the steady-state-parameter sensitivity effects on the rotor flux linkages, the ratio between the current and its command is derived as

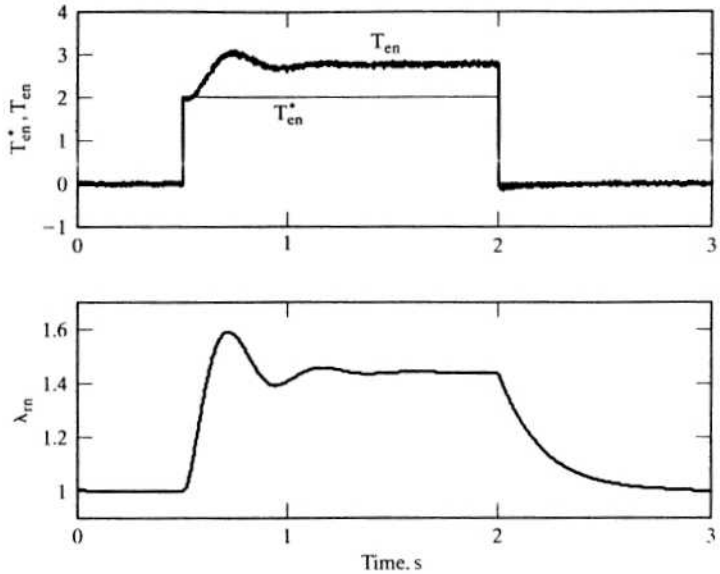


Figure 8.30 Simulation results for torque command at a stator temperature 100°C above ambient

$$\frac{i_f}{i_f^*} = \frac{i_s \cos\theta_T}{i_s^* \cos\theta_T^*} \tag{8.86}$$

The torque angle and slip speed are enforceable instantaneously, but not the current magnitude (because of the stator dynamics of the induction machine). The above relationship can be written in terms of rotor time constant and slip speed as

$$\frac{i_f}{i_f^*} = \frac{i_s}{i_s^*}\sqrt{\frac{1 + (\omega_{sl}T_r^*)^2}{1 + (\alpha\omega_{sl}T_r^*)^2}} \tag{8.87}$$

Considering the time lag between the stator current and its command, its transfer function is written as

$$\frac{I_f(s)}{I_f^*(s)} = \frac{1}{(1 + s\tau_c)}\sqrt{\frac{1 + (\omega_{sl}T_r^*)^2}{1 + (\alpha\omega_{sl}T_r^*)^2}} \tag{8.88}$$

from which the transfer function between the rotor flux linkage and its command is derived as

$$\frac{\lambda_r(s)}{\lambda_r^*(s)} = \beta\sqrt{\frac{1 + (\omega_{sl}T_r^*)^2}{1 + (\alpha\omega_{sl}T_r^*)^2}}\,\frac{(1 + sT_r^*)}{(1 + s\tau_c)(1 + s\alpha T_r^*)} \tag{8.89}$$

The rotor flux linkage, including the initial condition, is derived as

$$\lambda_r(s) = \beta\sqrt{\frac{1 + (\omega_{sl}T_r^*)^2}{1 + (\alpha\omega_{sl}T_r^*)^2}}\,\frac{\{(1 + sT_r^*)\lambda_r^*(s) - T_r^*\lambda_r^*(0)\}}{(1 + s\tau_c)(1 + s\alpha T_r^*)} + \frac{\alpha T_r^*}{(1 + s\alpha T_r^*)}\lambda_r(0) \tag{8.90}$$

For a step command of rotor flux linkages with a magnitude of λ_r^*, the time response is obtained from the above as

$$\lambda_r(t) = \beta\sqrt{\frac{1 + (\omega_{s1}T_r^*)^2}{1 + (\alpha\omega_{s1}T_r^*)^2}}\left[\lambda_r^* + \frac{1}{\tau_c - \alpha T_r^*}[(T_r^* - \tau_c)\lambda_r^* - T_r^*\lambda_r^*(0)]e^{-\frac{t}{\tau_c}} + T_r^*[\lambda_r^*(0) + \lambda_r^*(\alpha - 1)]e^{-\frac{t}{\alpha T_r^*}}\right] + \lambda_r(0)e^{-\frac{t}{\alpha T_r^*}} \tag{8.91}$$

The derivative term in the flux command path causes a large flux-producing component of the stator-current command to be generated, which is usually limited to rated value or slightly above it. Hence, to correspond with this reality, the solution for the response has to be modified by dropping the derivative term in the flux command path, resulting in the following solution:

$$\lambda_r(t) = \beta\sqrt{\frac{1 + (\omega_{s1}T_r^*)^2}{1 + (\alpha\omega_{s1}T_r^*)^2}}\left[\lambda_r^* + \frac{1}{\tau_c - \alpha T_r^*}\begin{matrix}[\tau_c\lambda_r^* - T_r^*\lambda_r^*(0)]e^{-\frac{t}{\tau_c}} \\ + T_r^*[\lambda_r^*(0) + \lambda_r^*]e^{-\frac{t}{\alpha T_r^*}}\end{matrix}\right] + \lambda_r(0)e^{-\frac{t}{\alpha T_r^*}} \tag{8.92}$$

Torque Response The torque-producing component of the stator-current response to a step command is evaluated; when combined with the rotor flux-linkages response, it gives the torque response. It is derived as follows. Like the transfer function between the rotor flux linkages and its command, the transfer function between the torque current and its command is derived as

$$\frac{I_T(s)}{I_T^*(s)} = \frac{\alpha}{(1 + s\tau_c)}\sqrt{\frac{1 + (\omega_{s1}T_r^*)^2}{1 + (\alpha\omega_{s1}T_r^*)^2}} \tag{8.93}$$

The time response due to a step command of torque current with a magnitude of i_T^* including an initial condition is derived as

$$i_T(t) = \alpha i_T^*\sqrt{\frac{1 + (\omega_{s1}T_r^*)^2}{1 + (\alpha\omega_{s1}T_r^*)^2}}[1 - e^{-\frac{t}{\tau_c}}] + i_T(0)e^{-\frac{t}{\tau_c}} \tag{8.94}$$

Then the torque response is written as

$$T_e(t) = \frac{3}{2}\frac{P}{2}\frac{L_m}{L_r}\lambda_r(t)i_T(t) \tag{8.95}$$

where the instantaneous values of rotor flux linkages and torque current are substituted from the equations derived in the above.

8.10.2 Parameter Sensitivity Effects on a Speed-Controlled Induction Motor Drive

The closure of an outer loop ensures that the electromagnetic torque, T_e^*, will be modified until the actual output torque is equal to the load torque in steady state, regardless of the parameter variations in the induction motor. The ratio of torque to its command value can be denoted by C. The relationship is expressed as

$$T_e = T_l = CT_e^* \tag{8.96}$$

The manner in which this relationship is enforced in the induction motor drive has consequences for many key variables of the drive. Insight into the drive's performance is possible by fixing the value of C and then evaluating the stator current, slip speed, and flux for a given load torque. The key variable to be solved for is the torque command for a given load torque. The relevant equations are derived next.

Let ω_{sl}^* be written from equation (8.57) as

$$\omega_{sl}^* = mT_e^* \tag{8.97}$$

where the following are defined:

$$m = K_{it}\frac{R_r^*}{(\lambda_r^*)^2} \tag{8.98}$$

$$x = \omega_{sl}^* T_r^* = hT_e^* \tag{8.99}$$

$$h = K_{it}\frac{R_r^* T_r^*}{(\lambda_r^*)^2} = \frac{4L_r^*}{3P(\lambda_r^*)^2} \tag{8.100}$$

By substituting equation (8.99) into equation (8.78), the ratio between the torque and its command is derived as

$$\frac{T_e}{T_e^*} = \alpha\beta\left[\frac{1 + x^2}{1 + \alpha^2 x^2}\right] = \alpha\beta\left[\frac{1 + (hT_e^*)^2}{1 + (\alpha h T_e^*)^2}\right] \tag{8.101}$$

Note that $T_e = T_l$ and that solving for T_e^* leads to the following cubic equation in T_e^*:

$$(T_e^*)^3 + (T_e^*)^2\left(-\frac{\alpha}{\beta}T_l\right) + T_e^*\left(\frac{1}{h^2}\right) + \left(-\frac{T_L}{\alpha\beta h^2}\right) = 0 \tag{8.102}$$

The roots of the equation are obtained by using one of the standard numerical methods. One real root constitutes the solution of equation (8.102); complex roots have no physical identity for the system considered. From T_e^*, the solutions for ω_{sl}^* and the rotor flux linkages are obtained from equation (8.99) as

$$\omega_{sl}^* = \frac{hT_e^*}{T_r^*} \tag{8.103}$$

and hence the ratio of rotor flux linkage to its command is given by

$$\frac{\lambda_r}{\lambda_r^*} = \beta\sqrt{\frac{1 + (hT_e^*)^2}{1 + (\alpha h T_e^*)^2}} \tag{8.104}$$

Peak stator current can be computed from h, T_e^*, and ω_{sl}^*.

8.10.2.1 Steady-state characteristics. Since the key drive variables are not related by simple expressions, as they are in the case of the torque mode of operation, a set of graphs is drawn to illustrate the effects of parameter sensitivity on the operation of the induction motor drive. All the graphs are drawn with load torque as the *x*-axis and the variables of interest on the *y*-axis. The major variables considered for study are the actual/command torque, actual/commanded rotor flux linkages, slip speed, and peak value of the stator current.

Figures 8.31 and 8.32 show the actual/commanded torque and actual/commanded rotor flux linkages for extreme values of $\alpha = 0.5$ and 1.5 and with a nominal value of 1, maintaining β at a value of 1. The torque and flux follow their command values for $\alpha = 1.0$ independent of operational speed. With decreasing α, the torque is less than its commanded value up to 0.75 p.u. of load torque, at which

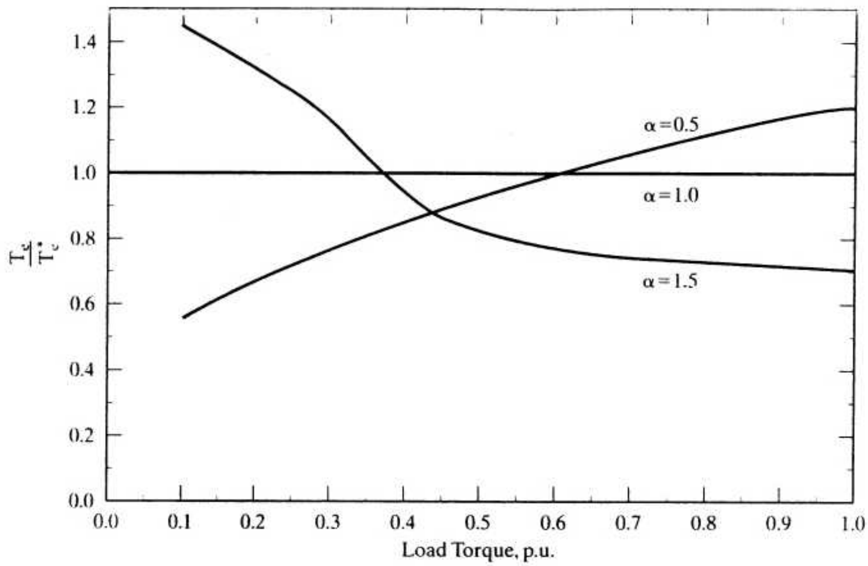


Figure 8.31 Torque over its commanded value vs. load torque for various values of α, with $\beta = 1.0$

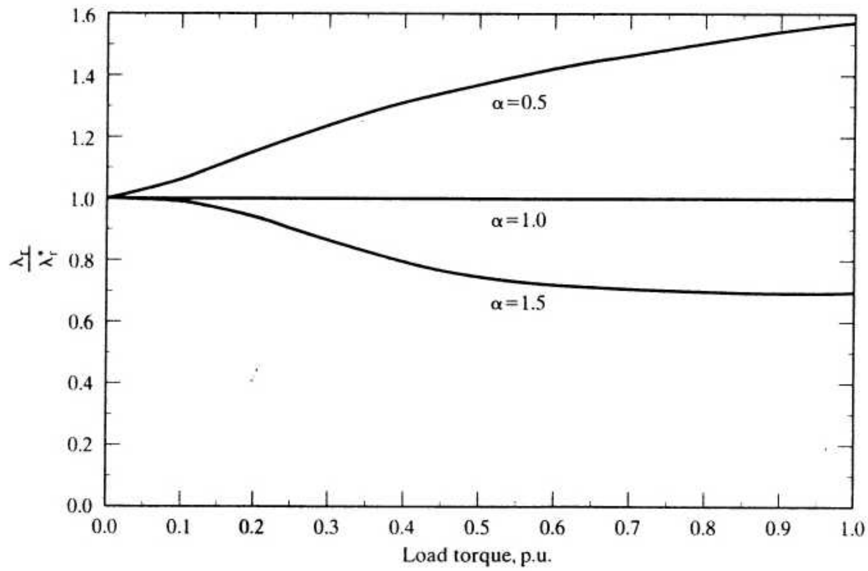


Figure 8.32 Actual/commanded rotor flux vs. load torque for various values of α, with $\beta = 1.0$

point the actual torque increases over its commanded value. The opposite trend in the behavior of the output torque is noticed with α greater than 1. Increasing α to greater than one radically decreases actual torque up to 0.75 p.u. of load torque, after which it remains at a constant level. At $\alpha = 0.5$, the actual flux increases monotonically with load torque. The saturation of the motor has a damping effect on this variable, and only modest increases in its magnitude result. The effect of saturation on torque and rotor flux is displayed in Figures 8.33 and 8.34. As expected, saturation (with $\beta = 0.8$) decreases the flux level by 20%, whereas the torque decreases with respect to its commanded value. The value of α is maintained at 0.5 in these graphs. Values of α greater than 1 have not been considered in these graphs. The reason lies in the fact that it is the lower values of α that are usually encountered in practice. Operation of the motor drive in the linear region of the B–H characteristics increases the value of β and, for a value of 1.2, the figures show the increase in both the flux and torque over their commanded values. The ratios of the actual to command values are plotted in order to appreciate the generalized nature of the characteristics. Values of 1.2 for the ratio of actual to commanded rotor flux at very low flux levels are a distinct possibility, but caution must be exercised in interpreting the flux, because the simulations are for the linear operating region.

Slip-speed characteristics are shown in Figures 8.35 and 8.36 for varying α and β. The slip speed is linear for $\alpha = 1$ but exhibits other trends for α greater than and less than 1. Defining the nominal value of a variable as that for $\alpha = 1$, if saturation is accounted for, in the case of $\alpha = 0.5$, the slip speed increases over its nominal value. There is a slight decrease in the slip speed for operation on the linear region of the B–H characteristics of the iron. Overall, for α less than 1, the slip speed is higher than the nominal value up to a reasonable fraction of full load, at which point

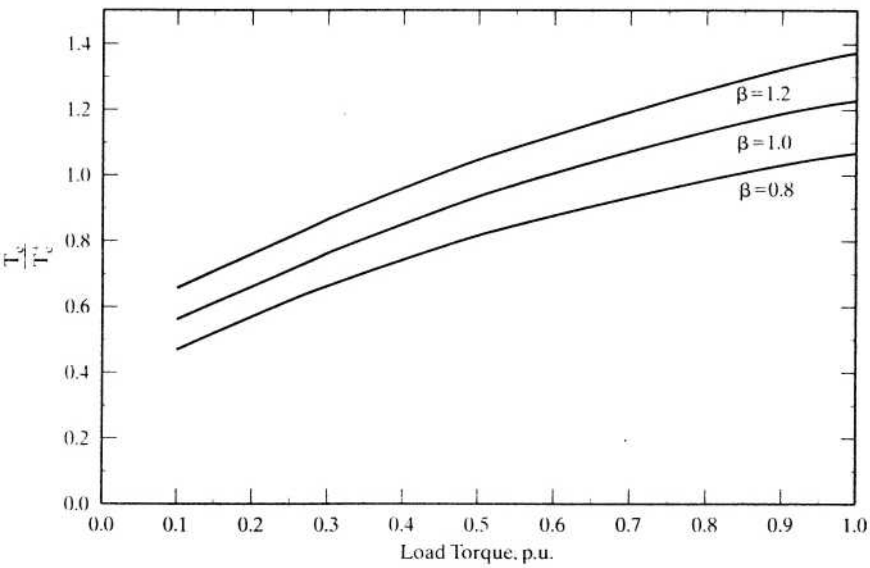


Figure 8.33 Torque over its commanded value vs. load torque for various values of β with $\alpha = 0.5$

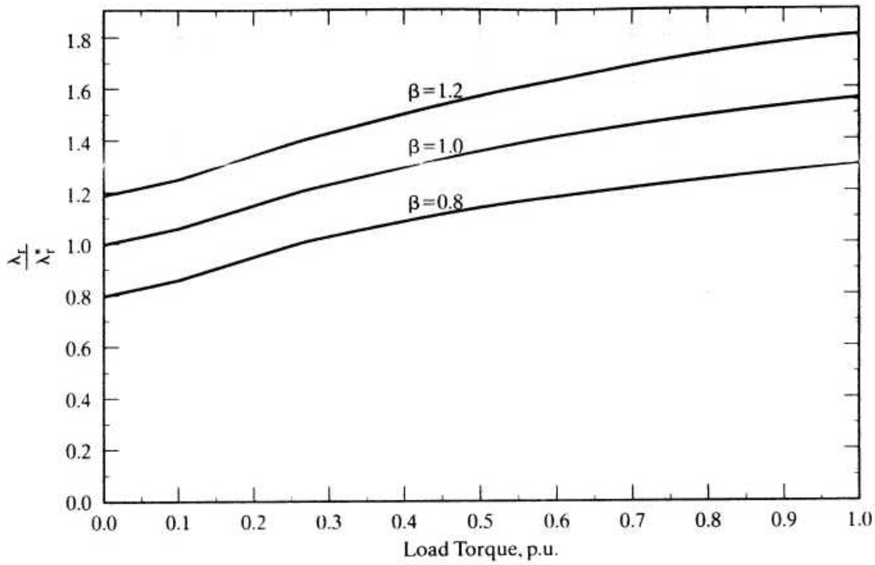


Figure 8.34 Actual/commanded rotor flux vs. load torque for various values of β, with $\alpha = 0.5$

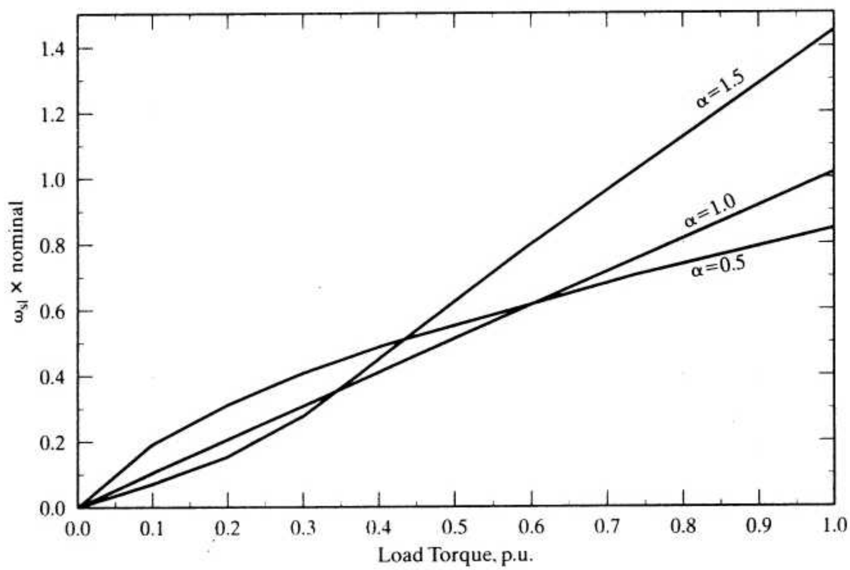


Figure 8.35 Slip speed vs. load torque for various values of α, with $\beta = 1.0$

it becomes slightly lower than the nominal value. This signifies that the rotor losses increase for load torques less than 0.8 p.u. In the field-weakening region of the drive, the losses increase for load torques higher than 0.4 p.u., as evidenced by the curve with α equal to 1.5. Notably, at $\alpha = 0.5$ and around 1 p.u. of load torque, the drive will have slightly lower losses than its nominal value and designed peak capacity, even when saturation is included in the computation.

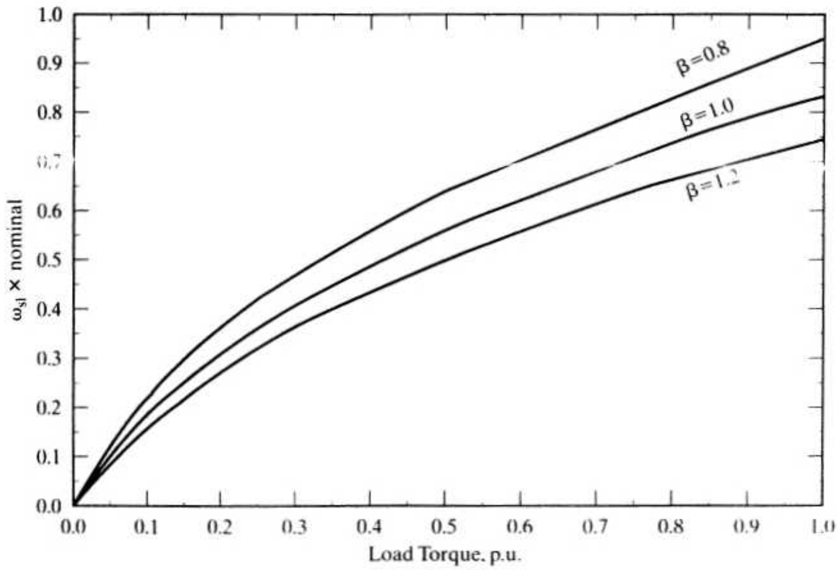


Figure 8.36 Slip speed vs. load torque for various values of β, with $\alpha = 0.5$

Peak values of the stator phase currents for discrete values of α and β are shown in Figures 8.37 and 8.38 against the load torque. A close resemblance to the slip-speed characteristics is evident. Saturation of the induction motor increases the input stator current as shown in Figure 8.38. The drive operation is usually confined to 1 p.u. torque and lower; within this region, the trend to have higher stator losses is evident for α less than 1. This is highly undesirable from the point of view of thermal rating and system efficiency.

8.10.2.2 Discussion on transient characteristics. It is well known that the stability of this drive system is not affected by machine-parameter sensitivity. Notably, the dampings of the oscillations in the flux and torque responses are dependent on the machine parameters. These oscillations in the flux and torque are not transmitted to the rotor shaft for two reasons:

1. the high bandwidth of the outer speed loop, which forces the torque demand to match the load torque in a very short time;
2. the filtering introduced by the moment of inertia of the induction motor and the load.

Saturation of the machine increases the field component of the stator current and decreases the torque component. Depending on the value of the magnetizing inductance, the field component of the stator current can vary from 0.25 to 0.7 p.u. Saturation curtails peak reserve electromagnetic torque, which, in turn, affects the dynamics of the drive, as reflected in increased acceleration and deceleration times.

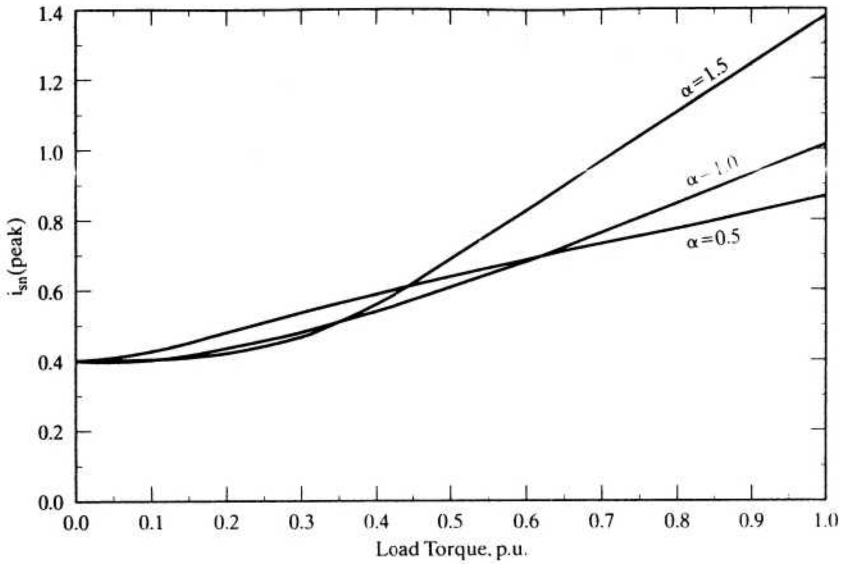


Figure 8.37 Peak stator current vs. load torque for various values of α, with β = 1.0

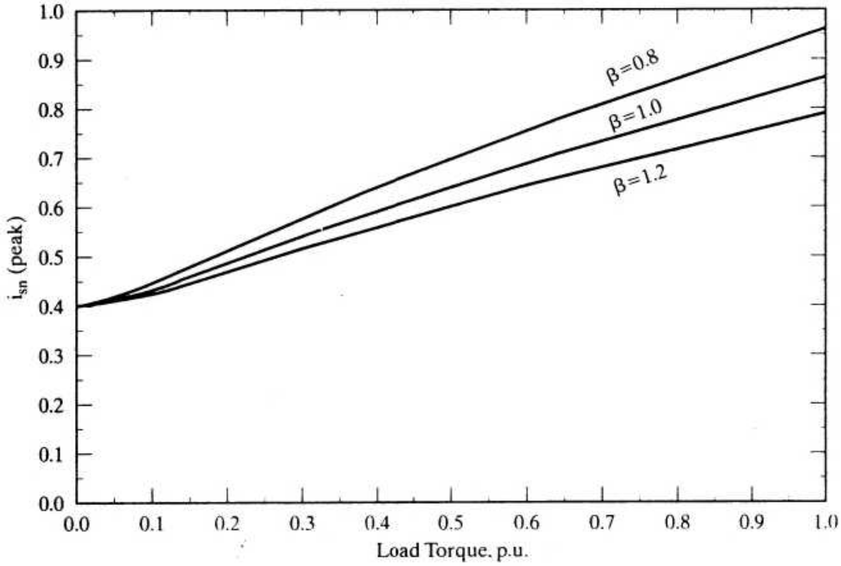


Figure 8.38 Peak stator current vs. load torque for various values of β, with α = 0.5

8.10.2.3 Parameter sensitivity of other motor drives. Ceramic permanent-magnet dc and synchronous motor drives have 20% less flux for a 100°C rise in temperature. A typical induction motor drive has almost the same sensitivity as a permanent-magnet motor drive, except that it is of positive sensitivity. The Samarium–Cobalt permanent-magnet machines have a sensitivity less by an order of magnitude compared to ceramic permanent-magnet motor drives.

Increasing temperature reduces the stator-current magnitude in the induction motor drive, whereas the drives with ceramic permanent magnets experience an increase in the armature/stator current; it is inversely proportional to the flux for a given torque. This factor not only contributes to an increase in the stator losses but also reduces the power output capability with a given inverter. For example, assuming a temperature rise of 100°C and an inverter rated for 2 p.u. power at ambient-temperature operating conditions, the output of a ceramic permanent magnet motor drive is reduced to 1.6 p.u. at 100°C because of a loss of 20% flux in the rotor. Consequently, there is a 20% reduction in the capacity of the motor drive. This is an interesting consequence; the induction motor drive has the opposite effect, enhancing power output under a similar situation. Including saturation effect, it could be conservatively stated that the capacity of the induction motor drive will not suffer from operation at temperatures higher than the ambient.

8.11 PARAMETER-SENSITIVITY COMPENSATION

The effects of mismatch between the induction motor and vector controller can be minimized by adapting the parameters in the vector controller to the actual motor parameters, at all times. This requires monitoring of the motor parameters, but direct monitoring is difficult while the drive is in operation. Several methods have come to the forefront to minimize the consequences of parameter sensitivity in indirect vector-control schemes. These are parameter-adaptation schemes based on one of the following strategies:

(i) direct monitoring of the alignment of the flux and the torque-producing stator-current component axes or real-time measurement of the instantaneous rotor resistance or both;

(ii) measurement of modified reactive power, measurement and estimation of rotor flux or the deviation of the field angle, or a combination of the rotor flux and the torque-producing component of the stator current. (An error in these measured variables is an indication of the amount of the parameter variation present in the induction motor.)

While strategy (i) can be classified as a direct scheme for parameter adaptation, strategy (ii) is an indirect parameter-adaptation scheme. Note that this classification is based on how the parameter-compensation signal is generated: by direct measurement of the parameter/alignment of the axes, or by indirectly monitoring the system for parameter variations as in strategy (ii). Most of the parameter-adaptation algorithms are themselves parameter-dependent. This particular aspect can cause significant error in the computation of the variables used in the parameter-compensation schemes. Besides such other critical factors as the speed of computation or method of measurement of the variables and complexity of the implementation, the error due to the parameter dependency of the algorithms could very well provide a yardstick to assess one scheme vis-à-vis other candidate schemes.

8.11.1 Modified Reactive-Power Compensation Scheme

A scheme using the modified reactive power, F, is discussed in this section, for parameter compensation of the indirect vector-controlled induction motor drive. The modified reactive power is defined by

$$F = -\frac{L_m}{L_r}\omega_s i_{ds}^e \lambda_r \tag{8.105}$$

and its command value is given as

$$F^* = -\frac{L_m^*}{L_r^*}\omega_s^* {i_{ds}^e}^* \lambda_r^* \tag{8.106}$$

and they are estimated from terminal voltages, phase currents, and other command variables. The modified reactive power command, F*, is obtained from the command variables of rotor speed, slip speed, flux-producing component of stator current, and rotor flux linkages. The computation of modified reactive power, F, can be shown to be, in terms of phase currents, their derivatives, and phase voltages,

$$F = \frac{2}{\sqrt{3}}\left[\left(v_{cs} - \frac{pi_{cs}}{K_{11}}\right)i_{as} - \left(v_{as} - \frac{pi_{as}}{K_{11}}\right)i_{cs}\right] \tag{8.107}$$

where

$$K_{11} = \frac{L_r}{(L_s L_r - L_m^2)} \tag{8.108}$$

The parameter K_{11} is dependent on motor inductances; hence, their variation will affect the computation of F and, in the final analysis, the parameter compensation. A change in the operating point will alter F* and, accordingly, F. Note that any parameter variation in the induction motor will change F and make it deviate from its expected or commanded value F*, indicating a change in the rotor time constant. The error between F* and F is amplified through a controller and a correction signal is obtained to correct the rotor time constant in the vector controller. This correction signal is made equivalent to the incremental slip speed required to compensate for the parameter variations in this particular implementation.

The implementation of this parameter-compensation scheme is shown in Figure 8.39. The error in the modified reactive power is converted into an incremental slip-speed signal and added to the slip-speed command of the vector controller. Figure 8.40 shows an experimental result from a laboratory prototype when the stator temperature was at 30°C. The motor drive is originally tuned at 20°C, where the parameters in the controller match those of the induction motor. With the temperature change, the torque characteristic has become nonlinear, as expected. In that situation, modified reactive power F is diverging from its command value, as shown in the plot. The parameter adaptation is achieved by closing the modified-reactive-power loop shown in Figure 8.39 by using estimated phase voltages and measured stator currents. Even without much tuning

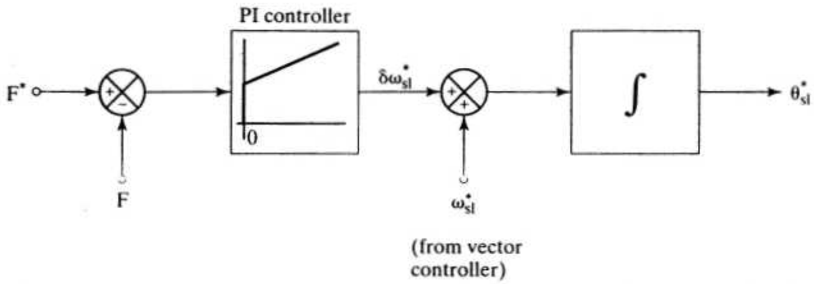


Figure 8.39 Block diagram of the parameter-adaptation with modified-reactive-power feedback control

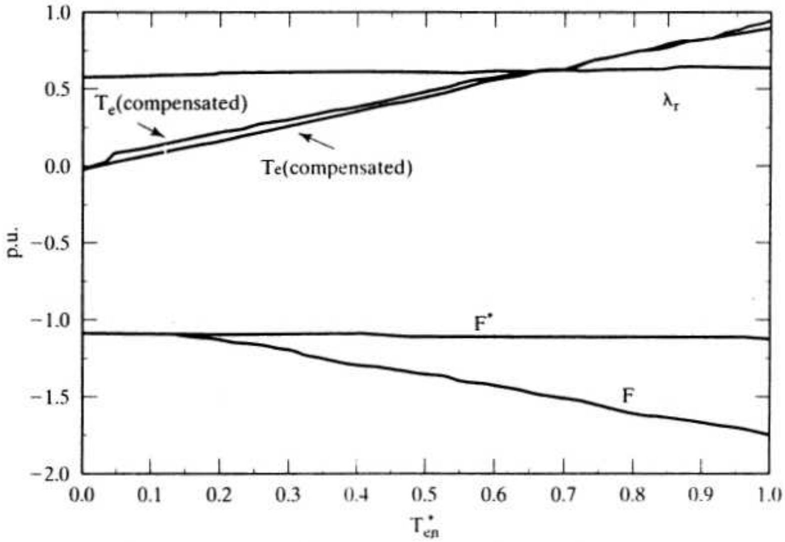


Figure 8.40 Experimental results of parameter adaptation in the induction motor drive

of the controller in the modified reactive power path, the torque characteristic has become linear. The parameter compensation is effective only from 2.5 N·m, i.e., 0.125 p.u. torque of the motor under parameter-compensated conditions, and that the estimated modified reactive power closely follows its command value is seen.

8.11.2 Parameter Compensation with Air Gap-Power Feedback Control

A parameter compensation scheme using air gap power is described in some detail in this section to overcome the dependence on inductances in the modified-reactive-power method. The basis of the scheme is that the air gap power in the machine is equal to its reference value when the controller and motor parameters match, but

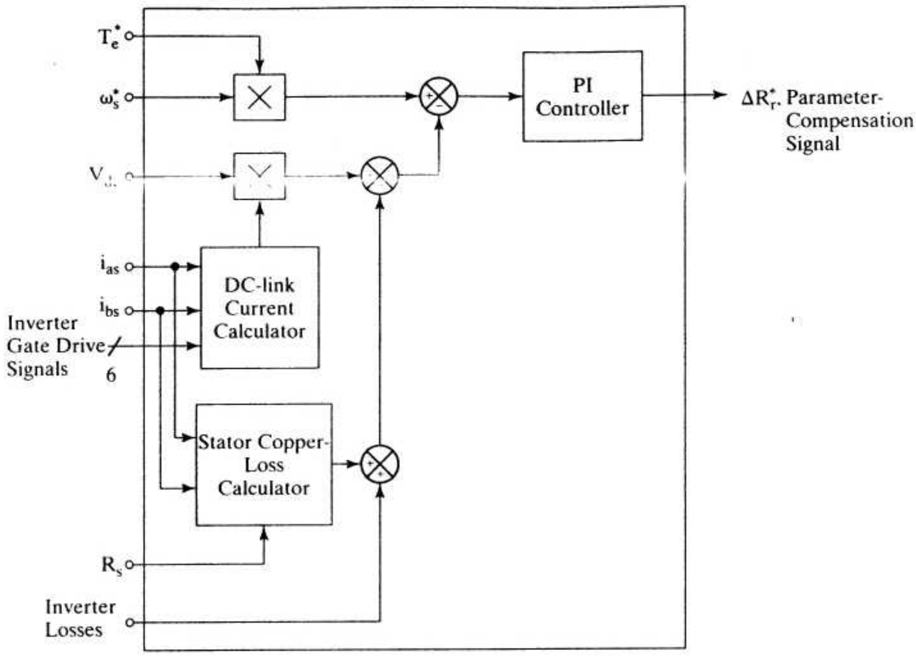


Figure 8.41 Parameter-compensation signal generation with air gap-power feedback control

develops an error in case of a divergence between the controller and motor parameters. This control scheme is shown in block-diagram form in Figure 8.41, with the dc-link voltage and stator currents as feedback variables. From the stator phase currents and inverter gate drive logic signals, the dc-link current can be reconstructed; when multiplied with the dc-link voltage, it gives the input power from the dc link. Subtracting the three-phase stator copper and inverter losses from the dc-link power provides the air gap power, P_a. The reference air gap power, P_a^*, is generated by the product of the torque and stator frequency references. The error between the reference and measured air gap power is amplified, then limited with a proportional-plus-integral controller to provide a parameter-compensation signal. Since this is a single signal and three parameters are involved in the vector controller, it is judicious to view this signal as an adjunct to the most sensitive parameter, i.e., rotor resistance in the controller. Hence, it is approximately identified as ΔR_r^*, which is summed with R_r^* in the block diagram of the slip-speed signal generator in the vector controller. The place of this parameter-compensation scheme in the overall indirect vector-controlled induction motor drive is shown in Figure 8.42 to appreciate the implementation aspect. Note that the parameter-compensation control is secondary to the indirect vector controller; accordingly, a lower priority is assigned during the execution of its control in software implementations.

8.11.2.1 Steady-state performance. The performance of this control scheme in steady state is explored for independent variations in rotor resistance and

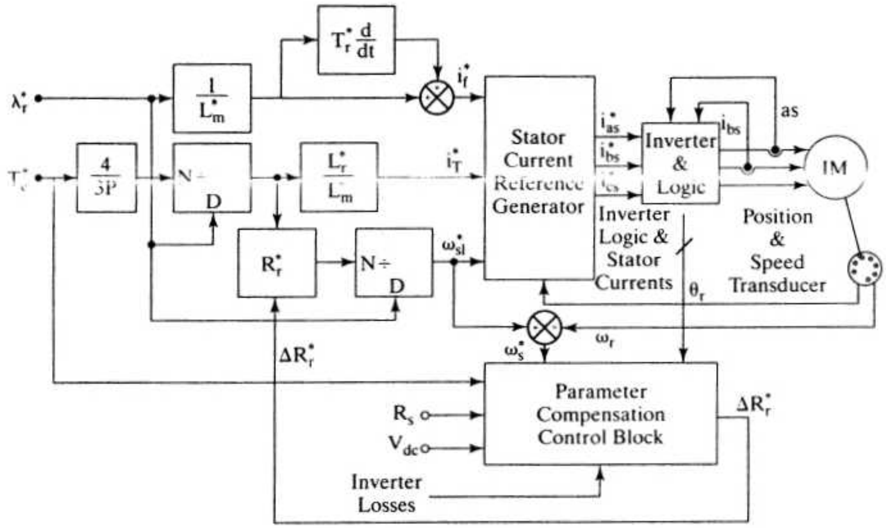


Figure 8.42 Parameter-compensated indirect vector-controlled induction motor drive

magnetizing inductance of the induction machine. The motor-phase rms current is obtained as

$$I_s^* = \frac{\sqrt{I_f^{*2} + I_T^{*2}}}{\sqrt{2}} \tag{8.109}$$

where

$$I_f^* = \frac{\lambda_r^*}{L_m^*} \tag{8.110}$$

$$I_T^* = \frac{4}{3P}\cdot\frac{L_r}{L_m}\cdot\frac{T_e^*}{\lambda_r^*} \tag{8.111}$$

This motor current, when applied with a stator frequency of ω_s given by

$$\omega_s = \omega_r + \omega_{s1}^* \tag{8.112}$$

where

$$\omega_{s1}^* = \frac{4}{3P}\cdot\frac{R_r^* T_e^*}{(\lambda_r^*)^2} \tag{8.113}$$

gives a rotor phase current, which is computed from the equivalent circuit as

$$I_r = \frac{\omega_s L_m}{\sqrt{\left(\frac{R_r}{s}\right)^2 + \omega_s^2(L_m + L_{1r})^2}}\cdot I_s^* \tag{8.114}$$

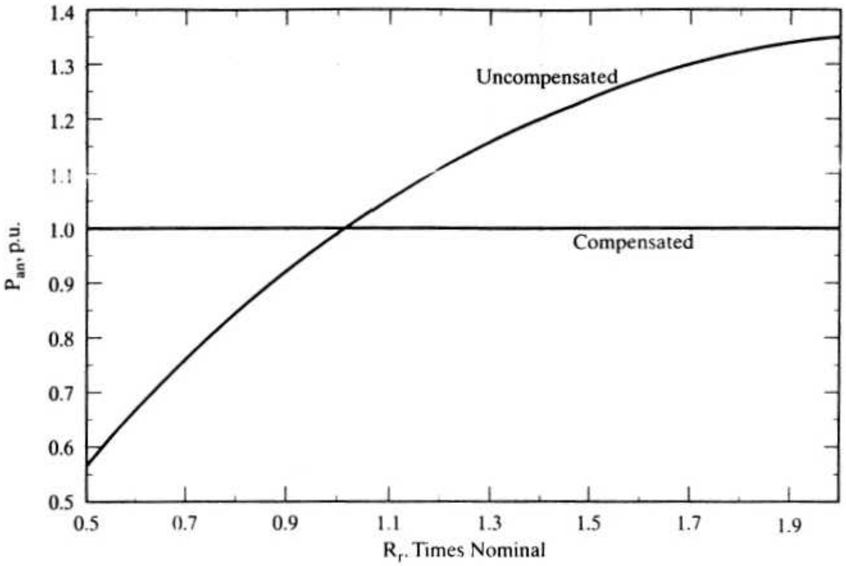


Figure 8.43 Effect of rotor-resistance variation on air-gap power at rated flux and torque commands with the rotor speed at rated value

The air gap power is then calculated as

$$P_a = 3I_r^2\frac{R_r}{s} \tag{8.115}$$

and the reference air gap power is given by

$$P_a^* = \frac{\omega_s T_e^*}{(P/2)} \tag{8.116}$$

where the parameters with asterisk indicate that they are the instrumented values in the controller.

These equations enable the determination of parameter sensitivity on air-gap power.

The effect of rotor-resistance variation in the machine from 0.5 to 2 times its nominal value, with nominal value set in the controller, on air gap power is shown in Figure 8.43. The flux and torque commands are maintained at rated values with the speed at rated value. The parameters of the machine are given in Example 8.4. Invariably, the air gap power is at rated value for the nominal value of rotor resistance that matches with the controller instrumented value. If the rotor resistance in the controller is made equal to the actual value in the machine that amounts to exact compensation, the air gap power remains at rated value, as is shown in the figure.

The effect due to variation in magnetizing inductance from 0.8 to 1.2 times nominal value is shown in Figure 8.44 for the same conditions as for the above. Both the variations produce a distinct error in air gap power from their reference value of

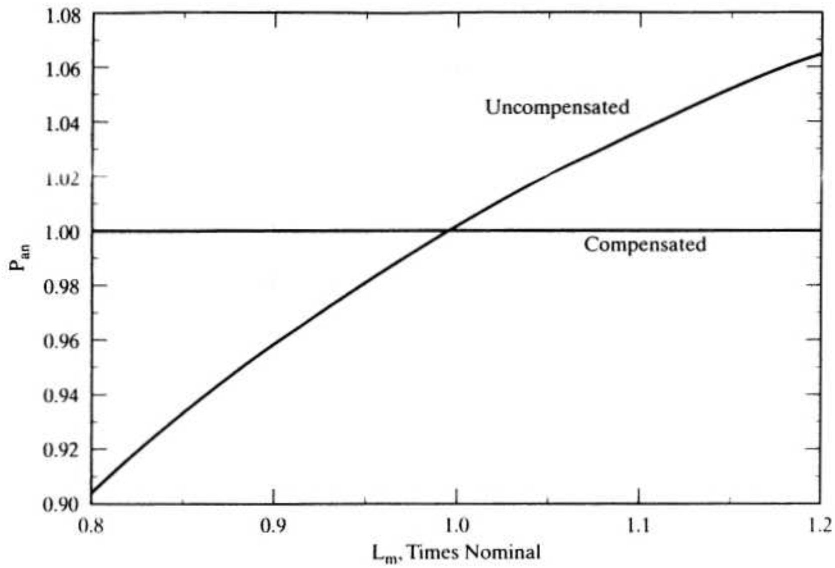


Figure 8.44 Effect of magnetizing-inductance variation on air gap power for rated flux and torque commands at rated speed

one p.u., thus clearly indicating that this air gap power error could be effectively employed to compensate for the parameter sensitivity of the indirect vector controller.

The magnetizing-inductance variation also is to be compensated in the vector controller, only in the slip-speed-signal generation, i.e., through the variation of the R_r^* value. In that case, the new value of R_r^* for variation of L_m can be calculated from the air gap equation with the following steps:

$$P_a = 3I_r^2 \frac{R_r}{s} = \frac{3\omega_{sl}^*(\omega_r + \omega_{sl}^*)(L_m I_s)^2 R_r}{R_r^2 + \omega_{sl}^{*2}(L_m + L_{lr})^2} \tag{8.117}$$

from which a quadratic equation in slip speed is obtained as

$$a\omega_{sl}^{*2} + b\omega_{sl}^* + c = 0 \tag{8.118}$$

where

$$a = 3(L_m I_s)^2 R_r - P_a(L_m + L_{lr})^2 \tag{8.119}$$

$$b = 3\omega_r R_r (L_m I_s)^2 \tag{8.120}$$

$$c = -P_a R_r^2 \tag{8.121}$$

Solving for the slip-speed command gives ω_{sl}^* as

$$\omega_{sl}^* = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{8.122}$$

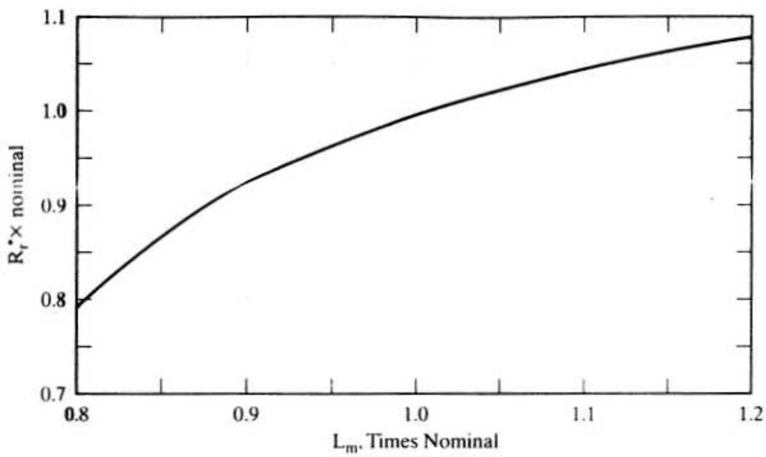


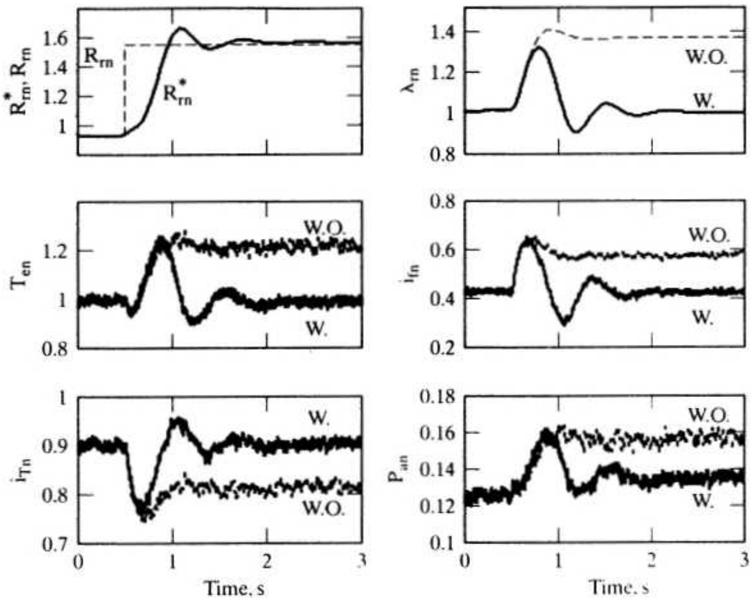
Figure 8.45 Variation of rotor resistance in the vector controller vs. magnetizing-inductance variation to deliver rated air gap power

from which the rotor resistance to be instrumented in the controller is obtained by

$$R_r^* = \left(\frac{3P}{4}\right)\frac{\omega_{sl}^*(\lambda_r^*)^2}{T_e^*} \tag{8.123}$$

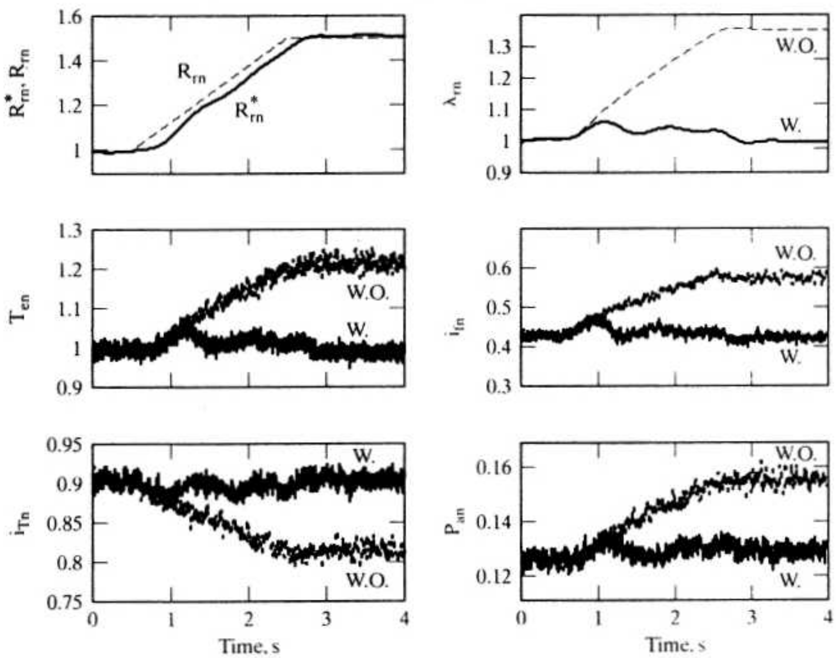
The rotor-resistance variation in the controller required to overcome the magnetizing inductance variation from 0.8 to 1.2 times nominal value is shown in Figure 8.45 for the same conditions discussed for the previous two figures. This figure verifies that the parameter compensation can be achieved through the slip-speed variation alone for all parameter changes in the vector controller.

8.11.2.2 Dynamic performance. The dynamic performance of the parameter-compensated vector-controlled drive with a step change in the rotor resistance from nominal to 1.5 times the nominal value is shown in Figure 8.46; that for a linear variation in R_r from nominal to 2 times nominal is shown in Figure 8.47. The responses demonstrate that the air gap-power feedback control for parameter compensation works and that the error in air gap power reduces to zero. Even that it takes 0.75 seconds to reach steady state is acceptable, because changes in rotor resistance are slowed down by the thermal time constant of the rotor. The steady-state error is not exactly zero; there is a window in the implementation of the air gap power error. If the error is within this window, the controller does not generate a correction signal. Steady-state error can be reduced to zero with a zero window error size, leading to oscillations in the dynamic response of the drive system. For the linear change in rotor resistance, which is more realistic in the motor, the parameter compensation is fairly swift and smooth, with oscillations only in the beginning as is evident from Figure 8.47.



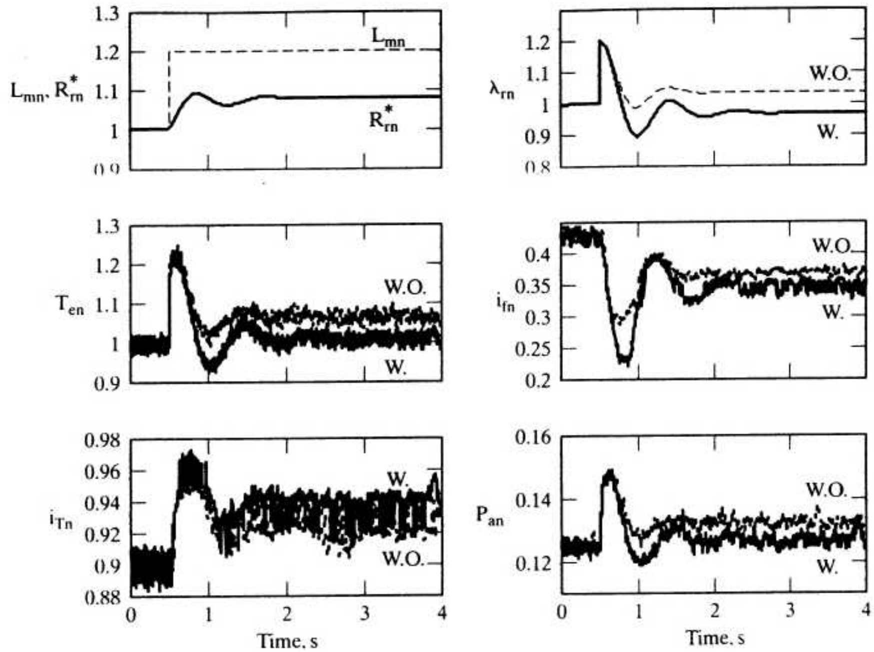
(W.O.–Without compensation, W.–With compensation.)

Figure 8.46 Plots for step change in rotor resistance



(W.O.–Without compensation, W.–With compensation.)

Figure 8.47 Plots for linear change in rotor resistance



(W.O.–Without compensation, W.–With compensation.)

Figure 8.48 Effects of increase in mutual inductance

A step change in mutual inductance by 20% likewise is effectively handled by this scheme. It is shown in Figure 8.48. Note that the rotor flux linkage is not properly compensated; that is the penalty for using the same compensation signal to counter three different parameter variations. The torque linearity is maintained intact in this parameter-compensation scheme.

8.12 FLUX-WEAKENING OPERATION

Principle of Flux Weakening The motor drive is operated with rated flux linkages up to a speed where the ratio between the induced emf and stator frequency, known as volts/Hz, with neglected stator impedance, can be maintained constant, i.e., at rated value. After a certain stator frequency, known as the base frequency, the volts/Hz ratio, and hence the stator flux linkage, will become lower than its rated value that is due to the fixed dc-link-voltage magnitude. The operation above the base speed necessarily involves the weakening of the flux, resulting in the reduction of torque for the same torque-producing component of the stator current. By coordinating the torque level for each operating speed, the power output can be maintained constant in the flux-weakening region, much as in the dc motor drive case.

This section covers the flux-weakening algorithms for stator (direct schemes) and rotor (indirect schemes) flux-linkage-controlled drive systems.

8.12.1 Flux Weakening in Stator-Flux-Linkages-Controlled Schemes

Consider the normalized stator voltage equations of the induction machine with flux linkages as variables. Neglecting the stator resistive-voltage drops and considering only the steady state, the equations are

$$v_{qsn}^e = R_s i_{qsn}^e + p\lambda_{qsn}^e + \omega_{sn}\lambda_{dsn}^e \cong p\lambda_{qsn}^e + \omega_s\lambda_{dsn}^e \tag{8.124}$$

$$v_{dsn}^e = R_s i_{dsn}^e + p\lambda_{dsn}^e - \omega_s\lambda_{qsn}^e \cong p\lambda_{dsn}^e - \omega_{sn}\lambda_{qsn}^e \tag{8.125}$$

$$v_{sn} = \sqrt{(v_{qsn}^e)^2 + (v_{dsn}^e)^2} = \omega_{sn}\sqrt{(\lambda_{qsn}^e)^2 + (\lambda_{dsn}^e)^2} = \omega_{sn}\lambda_{sn} \tag{8.126}$$

The stator flux-linkages phasor is derived as

$$\lambda_{sn} = \frac{v_{sn}}{\omega_{sn}} \tag{8.127}$$

In the case of the direct vector-control scheme, as with the indirect vector-control derivation, it may be assumed that the stator flux-linkages phasor is aligned with the d axis stator flux linkages. Thereby, the q axis stator flux linkages are forced to zero, with the resulting elegant torque expression in terms of the product of the stator flux-linkages phasor and the q axis current obtained as

$$\lambda_{qsn}^e = L_{sn} i_{qsn}^e + L_{mn} i_{qrn}^e = 0 \tag{8.128}$$

$$\lambda_{dsn}^e = L_{sn} i_{dsn}^e + L_{mn} i_{drn}^e = \lambda_{sn}^e \tag{8.129}$$

which, upon substitution in the torque expression, results as

$$T_{en} = \{i_{qsn}^e\lambda_{dsn}^e - i_{dsn}^e\lambda_{qsn}^e\} = i_{qsn}^e\lambda_{sn}^e \tag{8.130}$$

The air gap power is calculated as

$$P_{an} = T_{en}\omega_{sn} \tag{8.131}$$

By substituting for torque and flux linkages phasor from the previous equations, the normalized air gap power is derived as

$$P_{an} = T_{en}\omega_{sn} = i_{qsn}^e\lambda_{sn}^e\omega_{sn} = i_{qsn}^e\frac{v_{sn}^e}{\omega_{sn}}\omega_{sn} = v_{sn}^e i_{qsn}^e \tag{8.132}$$

If the q axis stator current is maintained constant, note that, in the flux-weakening region, the air gap power remains constant as the stator voltage phasor is maintained constant. The output power is the difference between the air gap power and the slip power. By keeping the slip constant, the output power becomes constant for a constant air gap power. Therefore, constant-power operation is obtained in the entire flux-weakening region of the stator-flux-linkages-controlled drive by forcing the stator flux linkages to be inversely proportional to stator frequency.

8.12.2 Flux Weakening in Rotor-Flux-Linkages-Controlled Schemes

To obtain the maximum speed of operation at rated power, a simple inverse variation of rotor flux linkages as a function of speed is not sufficient. During flux-weakening, the flux linkages are controlled in such a way that the stator-voltage phasor is within the admissible limit of the applied voltage. It is more natural, then, to control the stator flux linkages, which are directly related to the stator voltages, than the rotor flux linkages, which are indirectly related to the stator voltages. Note that high-performance drive systems are based on indirect vector control, in which the rotor flux-linkages phasor is varied as a function of the rotor speed. It is important, therefore, to have the flux weakening with the modified rotor flux linkages in indirect vector control, to effectively extend the range of speed of operation of the drive system.

Control of the rotor flux linkages affects the stator flux linkages, but not in a proportional manner. This results in the stator flux linkages being greater than the desired value, i.e., rated value, requiring stator voltages greater than rated values when the stator frequency exceeds the rated value. Since that could not be supplied from the limited dc link, constant-power operation is not possible to maintain in the flux-weakening mode. This is proven analytically in this section, to gain an insight into the process of the flux-weakening mode of operation in the indirect VCIM drives.

Consider the case where rotor flux is set inversely proportional to rotor speed. The relevant equations in steady state and in normalized units are

$$\lambda_{rn} = \frac{1}{\omega_{rn}} \tag{8.133}$$

$$I_{fn} = I_{frn}\lambda_{rn} \tag{8.134}$$

where λ_{rn} is rotor flux linkages in p.u., ω_{rn} is the rotor speed in p.u., I_{fn} is the field current in p.u., and I_{frn} is the rated field current in normalized units.

The stator equations in these variables are,

$$V^e_{qsn} = R_{sn}I_{Tn} + L_{sn}\omega_{sn}I_{fn} \tag{8.135}$$

$$V^e_{dsn} = R_{sn}I_{fn} - a_n\omega_{sn}I_{Tn} \tag{8.136}$$

where

$$a_n = \left(L_{sn} - \frac{L^2_{mn}}{L_m}\right) = \left(1 - \frac{L^2_{mn}}{L_m L_{sn}}\right)L_{sn} = \sigma L_{sn} \tag{8.137}$$

Neglecting the stator resistive-voltage drops, we have the stator voltages in steady state as

$$V^e_{qsn} \cong L_{sn}\omega_{sn}I_{fn} \tag{8.138}$$

$$V^e_{dsn} \cong -a_n\omega_{sn}I_{Tn} \tag{8.139}$$

and the stator-voltage phasor as

$$V_{sn} = \sqrt{(V^e_{qsn})^2 + (V^e_{dsn})^2} = \frac{L_{sn}}{L_{mn}}\omega_{sn}\sqrt{\lambda^2_{rn} + (\sigma L_{mn}I_{Tn})^2} \tag{8.140}$$

from which the rotor flux-linkages phasor is derived as

$$\lambda_{rn} = \sqrt{\left(\frac{L_{mn}}{L_{sn}}\frac{V_{sn}}{\omega_{sn}}\right)^2 - (\sigma L_{mn} I_{Tn})^2} \tag{8.141}$$

This equation demonstrates that the rotor flux linkage is not simply inversely proportional to the stator frequency; it is also strongly influenced by the torque-producing component of the current. The contrast between the rotor flux linkages in the indirect vector-control scheme and the stator flux linkages in the direct vector-control scheme is significant. In the latter case, the product of the stator flux linkages and stator frequency is constant for a fixed stator-voltages phasor. The product of the rotor flux linkages and stator frequency is not a constant for a fixed stator-voltage phasor in the indirect vector-control scheme. This fact implies that the air gap power cannot be maintained constant even if the rotor flux linkage is forced to be inversely proportional to stator frequency. The stator flux linkages increase under this circumstance, thus demanding a higher stator-voltage phasor, which may not be possible with a fixed dc-link voltage. The stator flux linkages increase under this circumstance, as shown in the following by rearranging equation (8.141) in terms of stator flux linkages, rotor flux linkages, and torque as

$$\lambda_{sn} = \frac{L_{sn}}{(L_{mn}\lambda_{rn})}\sqrt{bT_{en}^2 + \lambda_{rn}^4} \tag{8.142}$$

where

$$b = \sigma^2 L_{mn}^2 \tag{8.143}$$

This equation indicates clearly that the stator flux linkages are influenced by the rotor flux linkages and electromagnetic torque. As the electromagnetic torque is generated, the linear relationship between rotor and stator flux linkages is lost. Figure 8.49 shows the rotor flux linkages, electromagnetic torque, and voltage phasor versus rotor speed for a 1-p.u. stator current. Note that the stator voltage requirement exceeds 1 p.u. as the stator flux linkage increases. The key, then, to the control of flux-weakening operation is to maintain the stator flux linkages inversely proportional to the stator frequency. The constraint of a 1-p.u. current in steady state has to be further incorporated, as is given in the next section.

8.12.3 Algorithm to Generate the Rotor Flux-Linkages Reference

For the sake of simplicity, the stator resistances are neglected in the following derivations. This will introduce an error of 2 to 3% in the estimation of such variables as the torque and power in small machines, but much less error in large machines. Note that the following development makes it simple to incorporate the effects of the stator resistance if required for final tuning of the solution. To modify the rotor flux linkages to keep the stator voltage requirement to 1 p.u., a factor x is introduced, leading to

$$\lambda_{rn} = \frac{x}{\omega_{rn}} \tag{8.144}$$

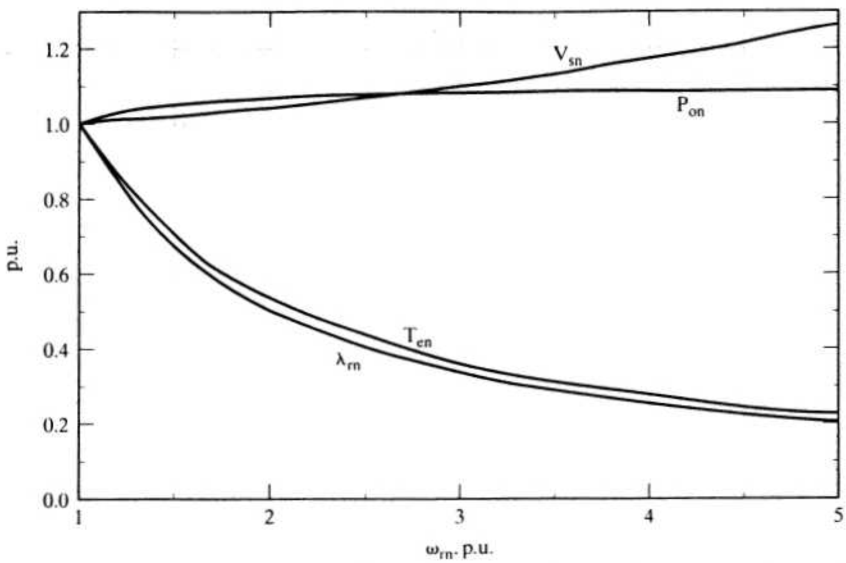


Figure 8.49 Performance characteristics with rotor flux inversely proportional to speed in the flux-weakening region

and

$$T_{en} = \frac{1}{\omega_{rn}} \tag{8.145}$$

and the torque-producing component of the stator current in terms of the above equations and rated values in normalized units is

$$I_{Tn} = I_{Trn}\frac{T_{en}}{\lambda_{rn}} = I_{Trn}\frac{1/\omega_{rn}}{x/\omega_{rn}} = \frac{I_{Trn}}{x} \tag{8.146}$$

where

$$I_{Trn} = \frac{I_{Tr}}{I_b} \tag{8.147}$$

and the rotor flux linkages in p.u. is

$$\lambda_{rn} = \frac{\lambda_r}{\lambda_{rb}} = \frac{L_m i_f}{L_m i_{fr}} = \frac{I_{fn}}{I_{frn}} \tag{8.148}$$

where

$$I_{frn} = \frac{I_{fr}}{I_b} \tag{8.149}$$

from which the field current is obtained as

$$I_{fn} = I_{frn}\lambda_{rn} = I_{frn}\frac{x}{\omega_{rn}} \tag{8.150}$$

The torque component of the stator current is

$$I_{Tn} = \frac{I_{Trn}}{x} = \sqrt{I_{sn}^2 - I_{fn}^2} = \sqrt{I_{sn}^2 - \frac{(xI_{frn})^2}{\omega_{rn}^2}} = I_{sn}\sqrt{1 - \left(\frac{xI_{frn}}{I_{sn}\omega_{rn}}\right)^2} \quad (8.151)$$

where the permissible stator current phasor is I_{sn}.

The synchronous speed is written as the sum of the rotor speed and slip speed, in normalized units:

$$\omega_{sn} = \omega_{rn} + \omega_{sln} = \omega_{rn} + \frac{\omega_{sl}}{\omega_b} \quad (8.152)$$

By substituting for the slip speed in terms of I_{Tn} and I_{fn}, the stator angular frequency is obtained as

$$\omega_{sn} = \omega_{rn} + \frac{R_r I_T}{L_r I_f}\cdot\frac{1}{\omega_b} \quad (8.153)$$

Substituting for the torque- and flux-producing components of stator currents, in terms of the variable x from equations (8.146) and (8.150), into the stator angular frequency equation gives

$$\omega_{sn} = \omega_{rn}\left\{1 + \frac{K_s}{x^2}\right\} \quad (8.154)$$

where the constant K_s is given by

$$K_s = \frac{R_r I_{Trn}}{\omega_b L_r I_{frn}} \quad (8.155)$$

Then the stator voltages are obtained, by substituting for the stator frequency into stator-voltage equations in steady state, as

$$V_{qsn}^e \cong L_{sn}\omega_{sn} I_{fn} = \left(\frac{x^2 + K_s}{x}\right) L_{sn} I_{frn} \quad (8.156)$$

$$V_{dsn}^e = -a_n\omega_{sn} I_{Tn} = -a_n\omega_{rn}\left(1 + \frac{K_s}{x^2}\right)\sqrt{I_{sn}^2 - \frac{x^2 I_{frn}^2}{\omega_{rn}^2}} \quad (8.157)$$

The magnitude of the stator-voltage phasor is given by

$$V_{sn} = \sqrt{(V_{qsn}^e)^2 + (V_{dsn}^e)^2} \quad (8.158)$$

By the combining of the last three equations, a polynomial in variable x is obtained:

$$d_1 x^6 + d_2 x^4 + d_3 x^2 + d_4 = 0 \quad (8.159)$$

where

$$d_1 = b_2^2 - a_n^2 I_{frn}^2 \quad (8.160)$$

$$d_2 = -v_{sn}^2 + 2K_s b_2^2 - 2K_s a^2 I_{frn}^2 + a_n^2\omega_{rn}^2 I_{sn}^2 \quad (8.161)$$

$$d_3 = b_2^2 K_s^2 - (aK_s I_{frn})^2 + 2K_s(a_n I_{sn}\omega_{rn})^2 \quad (8.162)$$

TABLE 8.5 Required rotor flux linkages vs. ω_{rn}

ω_{rn}, p.u.	x	$\lambda_{rn} = x/(1.03\omega_{rn})$ p.u.
1.0	1.030	1.000
1.5	1.013	0.656
2.0	0.990	0.480
2.5	0.958	0.372
3.0	0.919	0.297
3.5	0.870	0.241
4.0	0.809	0.196
4.5	0.733	0.158
5.0	0.636	0.123

$$d_4 = (a_n K_s \omega_{rn} I_{sn})^2 \tag{8.163}$$

$$b_2 = L_{sn} I_{frn} \tag{8.164}$$

Solving the polynomial equation for the roots of x will lead to the rotor flux linkages, and, together with the torque reference, the flux and torque components of the stator current and the slip speed can be evaluated. Note that the real root provides the solution for x; the other roots have no physical significance. The solution leads to the possible maximum torque and power output for a given stator voltage phasor and current phasor magnitude under the flux-weakening conditions. For the machine given in Example 8.4, the required rotor flux linkages to operate the drive system with 1-p.u. voltage and current magnitudes are given in Table 8.5 for various speeds.

At 1-p.u. speed, the rotor flux has to be 1 p.u., but x and λ_r have turned out to be 1.03 p.u. The additional 0.03 p.u. flux is due to the voltage that must have been dropped by stator resistance. Hence, to account for its neglect, the rotor flux linkage is modified by adjusting to 1 p.u. at 1-p.u. rotor speed. This is done by dividing all the ensuing rotor flux linkages by 1.03 in the flux-weakening region.

The rotor and stator flux linkages, torque, and power output versus rotor speed are shown in Figure 8.50 for this control with 1-p.u. voltage and current. The power output is not maintained at 1 p.u. beyond 2.5-p.u. speed. Increasing stator frequency rapidly increases the d axis stator voltage, necessitating a reduction in the q axis voltage by reducing the field current component of the stator current.

8.12.4 Constant-Power Operation

Constant-power operation is not obtainable over a wide speed range, such as 4 p.u. and above, in the illustrated case, but constant-power operation over a wide speed range is desirable in many applications, such as machine-tool spindles, electric vehicles, hand tools, and centrifuges. One of the solutions to this problem lies in having a reserve of dc-link voltage to meet the constant-power requirement. For example, the above illustration in Figure 8.49 shows that with 1.26-p.u. voltage, constant-power operation is achieved up to 5-p.u. speed with a 1-p.u. stator current. Several ways are available to increase the dc-link voltage or available voltage from the dc-link to the machine windings. Among them are increasing the dc-link voltage through a boost front-end rectifier, increasing the ac voltage by a step-up or tapped transformer, and changing the stator-

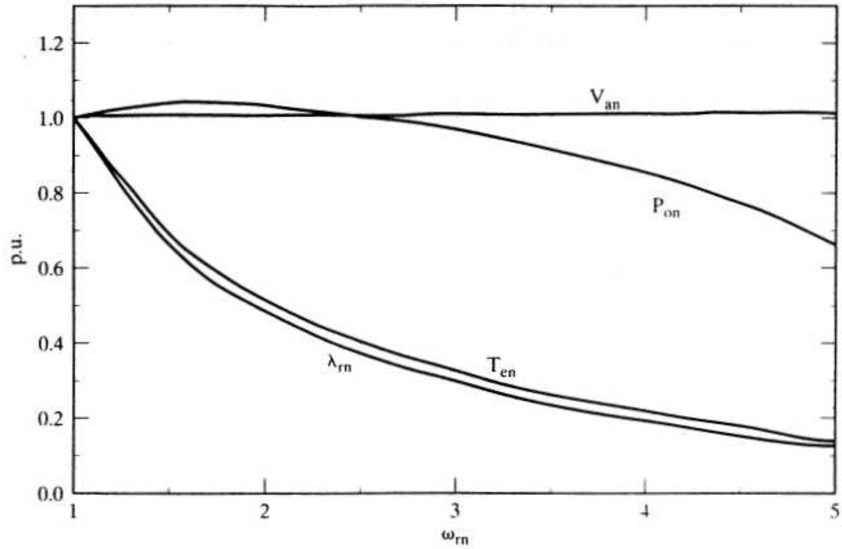


Figure 8.50 Performance characteristics of the vector-controlled induction motor drive with modified rotor flux weakening

windings connection from star to delta. While all of them are feasible, the solution using a step-up or tapped transformer is certainly not desirable: it increases the space requirement for the installation and also the cost. The other alternatives, using the boost front-end rectifier and changing the winding connections, are elegant solutions.

The boost front-end rectifier solves more than the one problem of varying the dc-link voltage; it also provides near-unity power factor while drawing near-sinusoidal currents, thus reducing the ac-line harmonic currents. Further, it makes the dc link independent of line variations compared to a diode-bridge rectifier in the front end. Such a solution is desired for emerging applications, to improve the line quality and to draw the minimum peak current and hence minimum kVA from the utility.

The star–delta change of winding connections enables higher voltage to be obtained for the machine phases, thus relieving the problem of insufficient voltage for flux weakening in the induction motor drives. Note that this method provides an inexpensive solution.

The solution methodologies suggested are suitable for general-purpose induction machines; no special design of the machines is required, and by providing a reserve of dc-link voltage, such as 20 to 30%, constant-power operation over a wide speed range could be obtained. The other approach is to coordinate the flux-weakening control with a motor of prespecified design parameters that would lend itself to working within the available dc-link voltage, yet be capable of providing the desired range of constant-power operation. This, then, would involve the coordination of motor design, which could be expensive for small-volume production but could turn out to be the most cost-effective for high-volume applications. In the light of increasing demand for ac drives as against dc drives, this solution becomes feasible for large-drive system manufacturers. The problem of determining the ideal machine parameters to provide constant-power operation over a wide range of speeds as a function of inverter variables along with other design constraints will be crucial in the product development of such drives in the future.

8.13 SPEED-CONTROLLER DESIGN FOR AN INDIRECT VECTOR-CONTROLLED INDUCTION MOTOR DRIVE

The principle, derivation, and implementation of decoupling nonlinear controllers for both the direct and indirect vector-control schemes made possible the independent control of flux and torque in the induction machine. In addition to torque control, speed control is required in a large number of applications. For speed regulation, usually an outer speed loop is closed in many applications. Then the design of the speed controller is of importance. An analytical approach using the transfer function is considered in the design of the speed controller. The vector controller transforms the induction motor drive into a linear system, even for large signals, when the flux linkages are maintained constant and hence resembles the separately-excited dc motor drive in all aspects, including in the development of the block diagram and hence in the synthesis of speed controller. This section contains the systematic development of the transfer-function derivation for the speed-controlled indirect vector-controlled induction motor drive. That similar derivations are possible for the direct vector controller goes without saying. Based on the transfer function, the speed controller is designed by using symmetric optimum method, but other methods could well be employed effectively, too. Use of symmetric optimum is made to maintain the uniformity of the speed-controller design for all ac and dc drive systems in this text.

8.13.1 Block Diagram Derivation

The block diagram of the indirect vector-controlled induction motor drive is derived in this section by developing the transfer functions of the various subsystems, such as the induction machine, inverter, speed controller, and feedback-transfer functions. By combining the subsystem block diagrams, the final block diagram of the induction motor drive is assembled, which has an overlap between the torque-current feedback loop and the induced-emf feedback loop. That overlap is overcome by block-diagram reduction techniques, making the inner current loop totally independent of the motor mechanical-transfer function; this approach lends itself to a simpler synthesis of the current controller, if need be.

8.13.1.1 Vector-controlled induction machine. To design the speed controller for the indirect vector-controlled induction motor drive, the key assumption of constant rotor flux linkages is made. Symbolically, the assumption leads to

$$\lambda_r = \text{a constant} \tag{8.165}$$

$$p\lambda_r = 0 \tag{8.166}$$

The stator equations of the motor are

$$v_{qs}^e = (R_s + L_s p)i_{qs}^e + \omega_s L_s i_{ds}^e + L_m p i_{qr}^e + \omega_s L_m i_{dr}^e \tag{8.167}$$

$$v_{ds}^e = -\omega_s L_s i_{qs}^e + (R_s + L_s p)i_{ds}^e - \omega_s L_m i_{qr}^e + L_m p i_{dr}^e \tag{8.168}$$

but from the vector controller, the following relationships of the rotor q and d axes flux linkages are made use of to recast the stator voltage equations as

$$i_{qr}^{e} = -\frac{L_m}{L_r} i_{qs}^{e} \tag{8.169}$$

$$i_{dr}^{e} = \frac{\lambda_r}{L_r} - \frac{L_m}{L_r} i_{ds}^{e} \tag{8.170}$$

Substitution of the rotor currents into the stator-voltage equations results in

$$v_{qs}^{e} = (R_s + \sigma L_s p) i_{qs}^{e} + \sigma L_s \omega_s i_{ds}^{e} + \omega_s \frac{L_m}{L_r} \lambda_r \tag{8.171}$$

$$v_{ds}^{e} = (R_s + \sigma L_s p) i_{ds}^{e} - \sigma L_s \omega_s i_{ds}^{e} + \frac{L_m}{L_r} p\lambda_r \tag{8.172}$$

where σ is the leakage coefficient. It is known that the flux-producing component of the stator current is constant in steady state, and that is the d axis stator current in the synchronous frames. Its derivative is also zero, giving the following:

$$i_f = i_{ds}^{e} \tag{8.173}$$

$$p i_{ds}^{e} = 0 \tag{8.174}$$

The torque-producing component of the stator current is the q axis current in the synchronous frames, given by

$$i_T = i_{qs}^{e} \tag{8.175}$$

Substituting these into the q axis voltage equation gives

$$v_{qs}^{e} = (R_s + L_a p) i_T + \omega_s L_a i_f + \omega_s \frac{L_m}{L_r} \lambda_r \tag{8.176}$$

where L_a is given by

$$L_a = \sigma L_s = \left(L_s - \frac{L_m^2}{L_r}\right) \tag{8.177}$$

Substituting for $\lambda_r = L_m i_f$ gives the q axis stator voltage in synchronous reference frames:

$$v_{qs}^{e} = (R_s + L_a p) i_T + \omega_s L_a i_f + \omega_s \frac{L_m^2}{L_r} i_f = R_s + L_a p i_T + \omega_s L_s i_f \tag{8.178}$$

The second stator equation is not required; the solution of either will yield i_T, which is the variable under control in the system. Now, the stator frequency is represented as

$$\omega_s = \omega_r + \omega_{sl} = \omega_r + \frac{i_T}{i_f}\left(\frac{R_r}{L_r}\right) \tag{8.179}$$

The electrical equation of the motor is obtained by substituting for ω_s from (8.179):

$$v_{qs}^e = (R_s + L_a p)i_T + \omega_r(L_s i_f) + \omega_{sl} L_s i_f = (R_s + L_a p)i_T + \omega_r(L_s i_f) + i_T\frac{R_r L_s}{L_r}$$

$$= \left(R_s + \frac{R_r L_s}{L_r} + L_a p\right)i_T + \omega_r L_s i_f \tag{8.180}$$

from which the torque-producing component of the stator current is derived as

$$i_T = \frac{v_{qs}^e - \omega_r L_s i_f}{R_s + \dfrac{R_r L_s}{L_r} + L_a p} = \frac{K_a}{(1 + sT_a)}\{v_{qs}^e - \omega_r L_s i_f\} \tag{8.181}$$

where

$$R_a = R_s + \frac{L_s}{L_r} R_r \tag{8.182}$$

$$K_a = \frac{1}{R_a} \tag{8.183}$$

$$T_a = \frac{L_a}{R_a} \tag{8.184}$$

From this block, which converts the voltage and speed feedback into the torque current, the electromagnetic torque is written as

$$T_e = K_t i_T \tag{8.185}$$

where the torque constant is defined as

$$K_t = \frac{3}{2}\frac{P}{2}\frac{L_m^2}{L_r} i_f \tag{8.186}$$

The load dynamics can be represented, given the electromagnetic torque and a load torque that is considered to be frictional for this particular case, as

$$J\frac{d\omega_m}{dt} + B\omega_m = T_e - T_l = K_t i_T - B_l \omega_m \tag{8.187}$$

which, in terms of the electrical rotor speed, is derived by multiplying both sides by the pair of poles:

$$J\frac{d\omega_r}{dt} + B\omega_r = \frac{P}{2} K_t i_T - B_l \omega_r \tag{8.188}$$

and hence the transfer function between the speed and the torque-producing current is derived as

$$\frac{I_T(s)}{\omega_r(s)} = \frac{K_m}{1 + sT_m} \tag{8.189}$$

where

$$K_m = \frac{P}{2}\frac{K_t}{B_t}; B_t = B + B_l; T_m = \frac{J}{B_t} \tag{8.190}$$

8.13.1.2 Inverter. The stator q axis voltage is delivered by the inverter with a command input that is the error between the torque-current reference and the torque-current feedback. This current error could be amplified through a current controller. The gain of the current controller is considered unity here, but any other gain can easily be incorporated in the subsequent development. The inverter is modeled as a gain, K_{in}, with a time lag of T_{in}. The gain is obtained from the dc-link voltage to the inverter, V_{dc}, and maximum control voltage, V_{cm}, as

$$K_{in} = 0.65\frac{V_{dc}}{V_{cm}} \tag{8.191}$$

The factor 0.65 here is introduced to account for the maximum peak fundamental voltage obtainable from the inverter with a given dc-link voltage. The interested reader is referred to Chapter 7 for details on this. The torque current error is restricted within the maximum control voltage, V_{cm}. The time lag in the inverter is equal to the average carrier switching-cycle time, i.e., half the period, and is expressed in terms of the PWM switching frequency as

$$T_{in} = \frac{1}{2f_c} \tag{8.192}$$

8.13.1.3 Speed controller. A proportional-plus-integral (PI) controller is used to process the speed error between the speed-reference and filtered speed-feedback signals. The transfer function of the speed controller is given as

$$G_s(s) = \frac{K_s(1 + sT_s)}{sT_s} \tag{8.193}$$

where K_s and T_s are the gain and time constants of the speed controller, respectively.

8.13.1.4 Feedback transfer functions. The feedback signals are current and speed, which are processed through first-order filters. They are given in the following.

(i) Current Feedback Transfer Function: Very little filtering is common in the current feedback signal; the signal gain is denoted by

$$G_c(s) = H_c \tag{8.194}$$

(ii) Speed-Feedback Transfer Function: The speed-feedback signal is processed through a first-order filter given by

$$G_\omega(s) = \frac{\omega_{rm}(s)}{\omega_r(s)} = \frac{H_\omega}{1 + sT_\omega} \tag{8.195}$$

where H_ω is the gain and T_ω is the time constant of the speed filter.

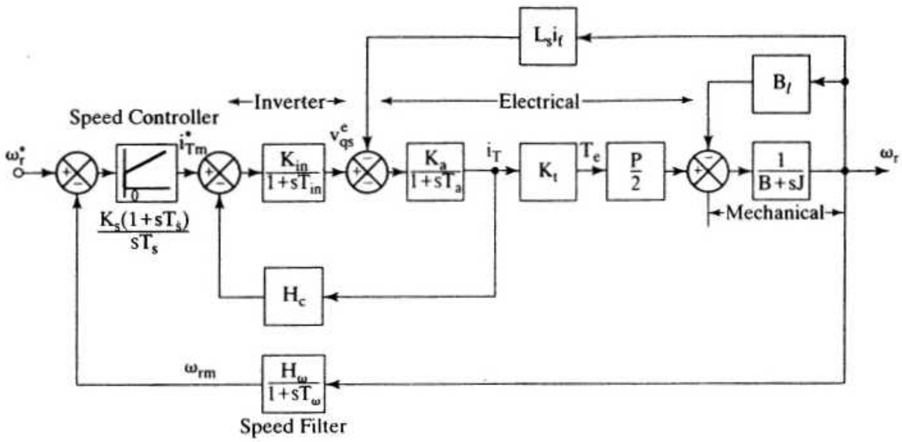


Figure 8.51 Block diagram of the vector-controlled induction motor with constant rotor flux linkages

The speed filter accepts the speed signal as input and produces a modified speed signal for comparison to the speed-reference signal, ω_r^*.

This completes the development of all the subsystems of the vector-controlled induction motor drive with constant rotor flux linkages. By incorporating equations (8.180), (8.181), (8.183), and from (8.191) to (8.195) with the mechanical impedance of the load, speed filter, speed controller, and i_T loop, the block diagram shown in Figure 8.51 is derived.

8.13.2 Block-Diagram Reduction

Further, the speed-feedback pickoff point for the electrical system can be moved to the i_T point resulting in the diagram shown in Figure 8.52(i) which can be further simplified as in Figure 8.52(ii), where the current closed loop transfer function

$$G_i(s) = \frac{K_aK_{in}(1 + sT_m)}{\{(1 + sT_{in})[(1 + sT_a)(1 + sT_m) + K_aK_b] + H_cK_aK_{in}(1 + sT_m)\}} \quad (8.196)$$

where the emf constant is given by

$$K_b = K_mL_si_f \quad (8.197)$$

8.13.3 Simplified Current-Loop Transfer Function

This third-order current transfer function, $\frac{i_{Tm}^*}{H_ci_T^*}$, can be approximated to a first-order transfer function as follows. T_{in} is usually negligible compared to T_1, T_2, and

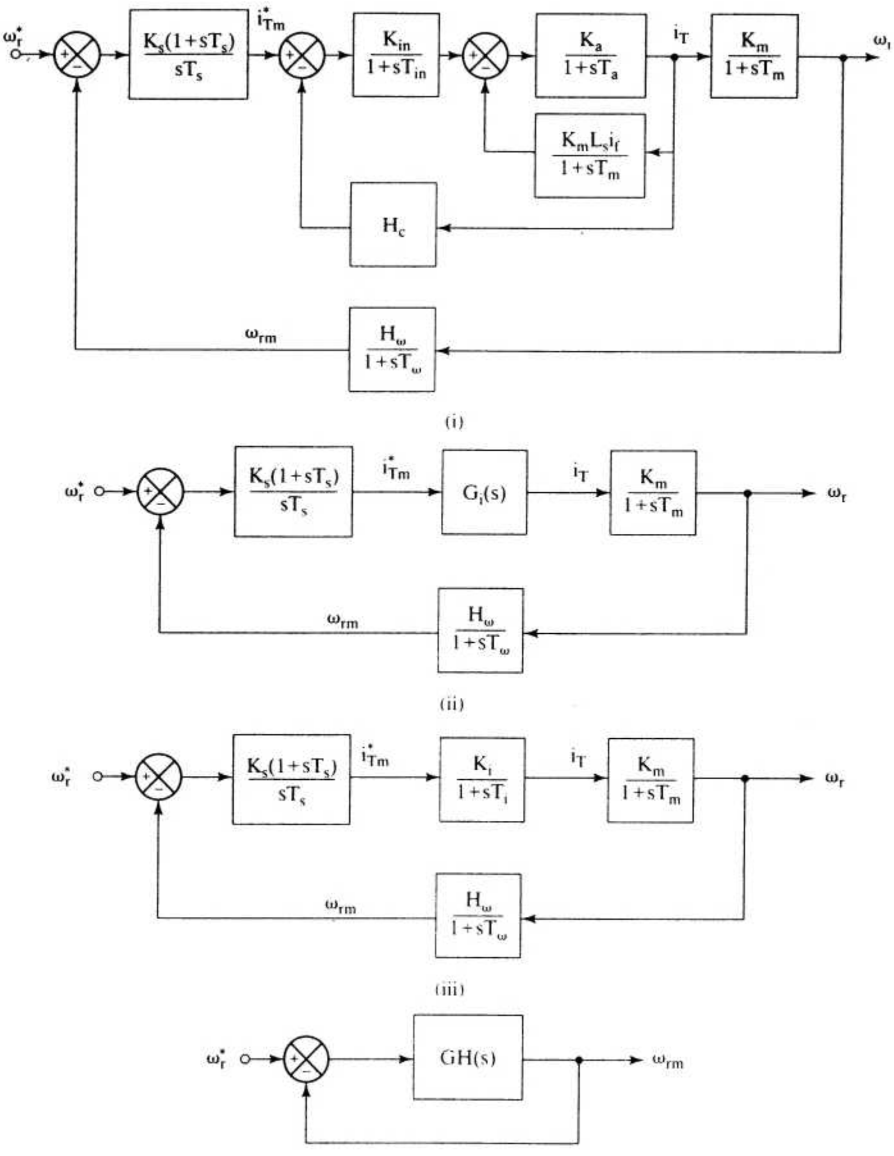


Figure 8.52 Block-diagram reduction of Figure 8.51

T_m, and, in the vicinity of the crossover frequency, the following approximations are valid:

$$1 + sT_{in} \cong 1 \tag{8.197}$$

$$(1 + sT_a)(1 + sT_{in}) \cong 1 + s(T_a + T_{in}) \cong 1 + sT_{ar}$$

where $T_{ar} = T_a + T_{in}$.

Substitution of these into $G_i(s)$ results in

$$G_i(s) = \frac{K_a K_{in}(1 + sT_m)}{(1 + sT_{ar})(1 + sT_m) + K_a K_b + H_c K_a K_{in}(1 + sT_m)} \tag{8.198}$$

which is written compactly as

$$G_i(s) = \frac{T_1 T_2 K_a K_{in}}{T_{ar} T_m} \cdot \frac{(1 + sT_m)}{(1 + sT_1)(1 + sT_2)} \tag{8.199}$$

where

$$-\frac{1}{T_1}, -\frac{1}{T_2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{8.200}$$

$$a = T_{ar} \cdot T_m$$

$$b = T_{ar} + T_m + H_c K_a K_{in} T_m$$

$$c = 1 + K_a K_b + H_c K_a K_{in}$$

The transfer function $G_i(s)$ is simplified by using the fact that $T_1 < T_2 < T_m$, and, near the vicinity of the crossover frequency, the following approximations are valid:

$$1 + sT_m \cong sT_m \tag{8.201}$$

$$1 + sT_2 \cong sT_2 \tag{8.202}$$

Substitution of these into $G_i(s)$ gives

$$G_i(s) = \frac{K_a K_{in} T_1}{T_{ar}} \cdot \frac{1}{(1 + sT_1)} = \frac{K_i}{(1 + sT_i)} \tag{8.203}$$

where K_i and T_i are the gain and time constants of the simplified current-loop transfer function, given by

$$K_i = \frac{K_a K_{in} T_1}{T_{ar}} \tag{8.204}$$

$$T_i = T_1 \tag{8.205}$$

The model reduction of the current loop is necessary to synthesize the speed controller. The loop transfer function of the speed is given then by the substitution of this simplified transfer function of the current loop, as shown in Figure 8.52(iii), and by combining all the blocks to obtain the final block diagram, as shown in Figure 8.52(iv).

8.13.4 Speed-Controller Design

The loop transfer function of the speed loop is given by

$$GH(s) \cong \frac{K_s}{T_s} K_g \frac{1 + sT_s}{s^2(1 + sT_{\omega i})} \tag{8.206}$$

where approximation $1 + sT_m \cong sT_m$ is made and the current-loop time constant and speed-filter time constant are combined into an equivalent time constant:

$$\begin{aligned} T_{\omega i} &= T_\omega + T_i \\ K_g &= K_i K_m \frac{H_\omega}{T_m} \end{aligned} \tag{8.207}$$

The transfer function of the speed to its command is derived as

$$\frac{\omega_r(s)}{\omega_r^*(s)} = \frac{1}{H_\omega}\left\{\frac{1 + sT_s}{1 + sT_s + \frac{T_s}{K_g K_s}s^2 + \frac{T_s T_{\omega i}}{K_g K_s}s^3}\right\} \tag{8.208}$$

and, by equating the coefficient of the denominator polynomial to the coefficient of the symmetric optimum function, K_s and T_s can be evaluated as given in Chapter 3. The symmetric optimum function for a damping ratio of 0.707 is given by

$$\frac{(1 + sT_s)}{1 + (T_s)s + \left(\frac{3}{8}T_s^2\right)s^2 + \left(\frac{1}{16}T_s^3\right)s^3} \tag{8.209}$$

from which the speed-controller constants are derived as

$$T_s = 6T_{\omega i} \tag{8.210}$$

$$K_s = \frac{4}{9}\frac{1}{K_g T_{\omega i}} \tag{8.211}$$

The proportional and integral gains of the speed controller are, respectively, then obtained as

$$K_p = K_s = \frac{4}{9}\frac{1}{K_g T_{\omega i}} \tag{8.212}$$

$$K_i = \frac{K_s}{T_s} = \frac{2}{27}\frac{1}{K_g T_{\omega i}^2} \tag{8.213}$$

The overshoot can be suppressed by canceling the zero with the addition of a pole $(1 + sT_s)$ in the path of the speed command. Consider the following example to test the validity of the various assumptions made in the derivation of the speed-controller design.

Example 8.8

Consider the induction motor given in example in Section 8.4, with the inverter and load parameters given below:

$$I_f = 6\text{ A}, f_c = 2000\text{ Hz}, B_t = 0.05, H_\omega = 0.05, T_\omega = 0.002, V_{cm} = 10\text{ V},$$
$$J = 0.0165\text{ kg} - \text{m}^2, V_{dc} = 285\text{ V}, H_c = 0.333\text{ V/A}$$

Calculate the speed-controller constants, and verify the validity of the assumptions in design.

Solution

$$R_a = R_s + R_r \cdot \frac{L_s}{L_r} = 0.457\ \Omega;\ K_a = \frac{1}{R_a} = 2.1885;\ L_a = L_s - \frac{L_m^2}{L_r} = 0.0047\text{ H}$$

$$T_a = \frac{L_a}{R_a} = 0.0104\text{ sec};\ T_m = \frac{J}{B_t} = 0.33\text{ sec};\ K_m = \frac{P}{2}\frac{K_t}{B_t} = 36.224$$

$$K_t = \frac{3}{2}\cdot\frac{P}{2}\cdot\frac{L_m^2}{L_r} I_f = 0.9056;\ \text{N}\cdot\text{m/A}\quad K_b = \frac{P}{2}\cdot\frac{K_t}{B_t}\cdot L_s I_f = 11.9672\text{ v/rad/sec};$$

$$T_{in} = \frac{T_c}{2} = \frac{1}{2f_c} = 0.00025\text{ sec};$$

$$K_{in} = 0.65\frac{V_{dc}}{V_{cm}} = 18.525\text{ V/V}$$

$$T_{ar} = T_a + T_{in} = 0.01065\text{ sec};\quad T_1 = 0.00074\text{ sec};\quad T_2 = 0.1173\text{ sec}$$

Approximated current loop:

$$K_i = \frac{K_{in}}{R_a} = 2.8708;\ T_i = T_1 = 0.00074\text{sec}$$

Speed controller:

$$K_g = \frac{K_i K_m H_\omega}{T_m} = 104.1;\ T_{\omega i} = T_\omega + T_i = 0.00274\text{ sec};\quad T_s = 6T_{\omega i} = 0.0164\text{ sec};$$

$$K_s = \frac{4}{9K_g T_{\omega i}} = 1.5605$$

Proportional gain, $K_{ps} = K_s = 1.5605$

Integral gain, $K_{is} = \dfrac{K_s}{T_s} = 95.0655$

With these calculated values, the following transfer functions are obtained:

Exact current loop transfer function, $G_i(s) = \dfrac{G_{in}(s)G_{12}(s)}{1 + H_c G_{in}(s)G_{12}(s)}$

where

$$G_{in}(s) = \frac{K_{in}}{1 + sT_{in}};\ G_{12}(s) = \frac{G_1(s)}{1 + G_1(s)G_2(s)};\ G_1(s) = \frac{K_a}{1 + sT_a};\ G_2(s) = \frac{K_b}{1 + sT_m}$$

Simplified current-loop transfer function, $G_{is}(s) = \dfrac{K_i}{1 + sT_i}$

$$\text{Exact speed-loop transfer function, } G_{\omega e}(s) = \frac{\omega_r(s)}{\omega_r^*(s)} = \frac{G_{\omega f}(s)}{1 + G_{\omega f}(s)G_\omega(s)}$$

where

$$G_{\omega f}(s) = \frac{K_m}{(1 + sT_m)} \cdot \frac{K_s(1 + sT_s)}{sT_s} \cdot G_i(s) : \; G_\omega(s) = \frac{H_\omega}{1 + sT_\omega}$$

Simplified speed-loop transfer function: Obtained from simplified current loop-transfer function and given by

$$G_{\omega s}(s) = \frac{G_f(s)}{1 + G_f(s)G_\omega(s)}$$

where

$$G_f(s) = \frac{K_m}{(1 + sT_m)} \cdot \frac{K_s(1 + sT_s)}{sT_s} \cdot \frac{K_i}{(1 + sT_i)}$$

The smoothed speed-loop transfer function is obtained by introducing a pole of $-\frac{1}{T_s}$ in the forward path of the speed reference, resulting in approximate and exact smoothed speed-loop transfer functions:

$$\text{Approximate: } G_{s\omega s}(s) = \frac{G_{\omega s}(s)}{1 + sT_s}$$

$$\text{Exact: } G_{s\omega e}(s) = \frac{G_{\omega e}(s)}{1 + sT_s}$$

All these transfer functions are computed; their gain and phase plots are given in Figures 8.53(i), (ii), and (iii). Even though there seems to be a significant discrepancy between the gains of the simplified and the exact current-loop transfer functions, note that their phases are identical in the frequency range of interest. The discrepancy due to the approximation of the current loop from third order to first order has hardly affected the accuracy of the speed-loop transfer functions, as is clearly seen from the gain and phase plots. This justifies the assumptions made in various approximations.

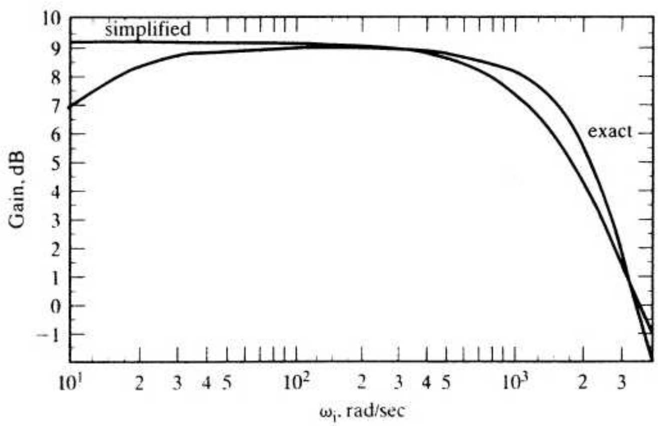


Figure 8.53(i) Frequency response of complete and simplified current loops

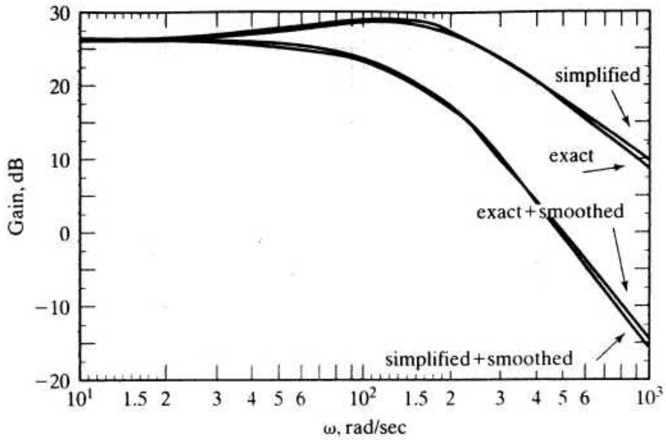


Figure 8.53(ii) Comparison of speed-loop frequency responses with and without smoothing with complete and simplified current loops

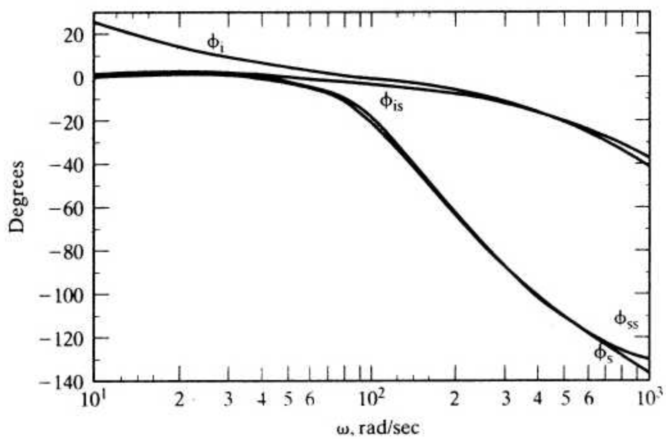


Figure 8.53(iii) Phase plots for current- and speed-loop transfer functions

Note that φ_{is} is the simplified current-loop phase, φ_i is its exact phase, φ_{ss} is the simplified, and φ_s is the exact phase of speed-loop transfer functions. The phase plot for the smoothed speed-loop transfer function is not shown; it is evident from φ_{is} and φ_s.

8.14 PERFORMANCE AND APPLICATIONS

The vector-controlled induction motor drive has delivered high performance. Some of its key performance achievements are given in the following:

(i) more than 6 kHz of torque bandwidth;

(ii) 200-Hz small-signal speed-loop bandwidth;

(iii) precise and smooth zero-speed operation;

(iv) trouble-free four-quadrant operation;
(v) fast torque and speed reversals.

Such performance figures have enabled these motor drives to find applications in

(i) machine-tool servos and spindles;
(ii) robotic actuators;
(iii) punch presses;
(iv) electric traction and propulsion;
(v) rolling mills;
(vi) paper and pulp drives;
(vii) centrifuges.

An illustrative example of the vector-controlled induction motor drive for centrifuges is given in the next section.

8.14.1 Application: Centrifuge Drive [58]

Centrifuges are used to separate the granular and crystalline parts from saturated solutions by spinning them at controlled speeds for set times. For example, a sugar centrifuge application is described with its speed and torque requirements over a cycle of operation. The sugar centrifuge has a basket, which is connected to the motor shaft through an elastomer coupling, as shown in Figure 8.54. The supersaturated sugar solution, with a thick suspension of sugar crystals, will be poured into the basket when the centrifuge is spinning at 250 rpm. After charging with the solution, the centrifuge basket is accelerated to a speed of 1200 rpm. This speed is maintained until the sugar solution is forced through the perforations in the bottom of the basket. This operation is known as the spin cycle. During spin, note that the motor is delivering friction and windage losses and hence requires the development of a very small electromagnetic torque. During the spin cycle, the sugar crystals stick to the walls of the basket and become dry. Before the removal of the sugar crystals, the basket has to be stopped. To stop it in a short time and to recover the kinetic energy stored in the basket, a maximum braking torque is applied to the centrifuge. The removal of dry sugar crystals is effected by spinning the basket in the reverse direction at 35 rpm and requires nearly 0.5 p.u. of torque to loosen the crystals. The sugar crystals are discharged through the bottom. A complete cycle of operation is shown in Figure 8.55. From the operational cycle it is evident that, to improve productivity, the acceleration and deceleration have to be swift, which requires peak torque capability from standstill to maximum speed. The deceleration can be smooth and fast only if it is done under regenerative conditions. Further, a controlled speed reversal and low-speed operation with 0.5 p.u. of torque to loosen the sugar crystals for final discharge to packaging requires a four-quadrant drive with very high torque and speed bandwidths. The power requirement can be up to 250 kW for the centrifuges: the baskets are 1219 mm (48 inches) in diameter and 762 mm (30 inches) deep and weigh considerably with the charge in them. From these requirements, the autosequentially commutated current-source inverter (ASCI)-driven induction motor emerges as the strong choice for this application, provided that the following modifications are incorporated in the design of the inverter control.

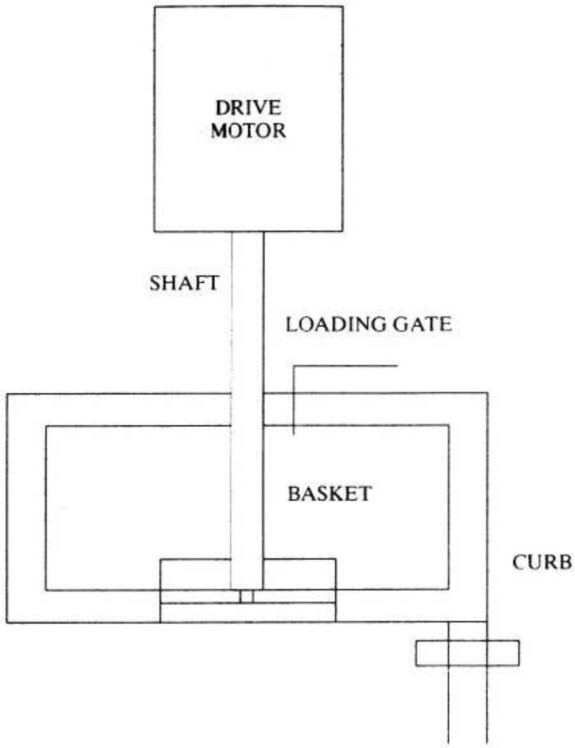


Figure 8.54 Sugar centrifuge schematic

- To reduce the torque pulsations at low speeds, PWM of the current pulses has to be provided for.
- For faster torque response, vector control is required to overcome the oscillatory responses and to develop the maximum torque per ampere input.

Note that the ASCI induction motor drive is superior to dc drives, because it can be operated under hazardous environment, having no commutator or brushes. Compared to other inverter drives, ASCI is cost effective, highly reliable, and simpler in construction.

8.15 RESEARCH STATUS

The present research is directed to the formulation and experimental verification of parameter-compensation schemes, speed and position sensorless schemes in the vector-controlled induction motor drives. Given the variety of parameter-compensation schemes available, effort is made to compare and contrast and to choose the best of them. All the compensation schemes require feedback of many machine variables. Inexpensive means of acquiring them remains to an extent a seri-

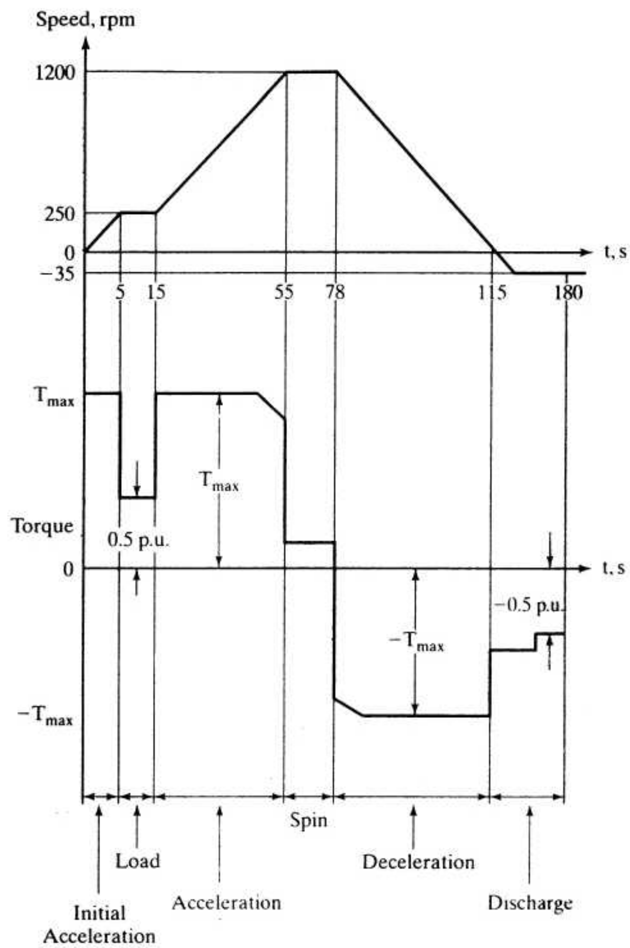


Figure 8.55 A complete cycle of operation in a sugar centrifuge

ous problem in the implementation of parameter-compensation schemes. This is being realized, and some efforts are being made in this direction.

Selftuning algorithms are emerging to match a vector controller to an induction motor with unknown parameters. This is the best approach to overcome the secondary problem of parameter compensation.

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8.17 DISCUSSION QUESTIONS

1. The command variables i_f^* and i_T^* are in synchronously rotating reference frames in the vector control scheme. Is it possible to have them in rotor reference frames? Are there any special advantages in doing so?
2. The field angle θ_f could be obtained by using flux-sensing coils. Discuss the measurement scheme in a block-diagram form. Discuss the situation when the stator windings are fed dc currents.
3. Using flux-sensing coils and output signals, devise a scheme to measure the rotor flux linkages and electromagnetic torque. (Hint: Reference 39.)
4. Vector control is the instantaneous control of the stator-current phasor. This implies that the performance of the motor drive is very much dependent on the response of the inverter. Discuss the impact of switching speed on the transient response of the motor drive.
5. In high-performance drives, two types of current control are commonly devised. They are hysteresis control (or bang-bang control) and PWM control. Discuss the impact of these controllers on the performance of the motor drive.
6. Which is the most significantly changing parameter in the induction motor and why? In that case, identify a simple and inexpensive parameter-compensation scheme.

7. A flux or torque feedback control is sufficient to overcome the effects of parameter sensitivity of the induction motor drive. Justify this statement.
8. Is there a method to differentiate among the various parameter-compensation schemes? Explain the basic principle of such a method. Discuss the merits and demerits of the method. (Hint: Reference 28.)
9. Discuss the high performance of vector-controlled induction motor drives *vis-à-vis* dc motor drives and permanent-magnet synchronous motor drives.
10. There are many variations of vector control based on the same principle of instantaneous control of the stator-current phasor. Summarize the key variations, their merits, and demerits. (Hint: Reference 37.)
11. Prove that position or speed signal is not obtainable from the fundamental model with sinusoidally distributed windings and with no saliency in the rotor structure.
12. Saliency on the rotor can be introduced by any one of the following:
    (i) varying the rotor copper fill;
    (ii) varying the rotor slot opening;
    (iii) varying the rotor slot height;
    (iv) varying the tooth lip thickness.

    Will these methods lend themselves to realistic implementation in an actual drive system? Discuss the merits and demerits of use of saliency for position detection.
13. Rotor-slot harmonics can be determined from the stator-current spectrum. This then has indirectly the position and speed information. This signal is extracted by passing through a band-pass filter and extracted with a phase-locked loop. At low speeds, this signal has a large time constant and is not usable for high dynamic performance at zero and low speeds. Justify this statement.
14. Direct self-control does not rely on position sensor feedback. Is this correct?
15. For high performance, stator or rotor flux-linkages-position information is required in the vector controller. Explain why?
16. How does direct self control get its position information?
17. Stator resistance is the only machine parameter that direct self control is dependent on for its implementation. Stator-current error can be used to adapt the stator resistance. What other signals can be used for stator-resistance adaptation?
18. Air gap power feedback is ineffective for tracking stator-resistance variations. Explain why.
19. A stator-flux-linkages-based vector controller does not give a simple decoupling of flux and torque channels, as in the indirect vector controller. Prove this statement.
20. When the stator flux-linkages position is unavailable or has significant error at zero and low speeds, how can the direct-self-control-based induction motor operate? Will there be a degradation of performance, and, if so, how can it be averted?
21. In the torque and flux feedback loops, the fundamental components of currents and voltages or their actual values, including the harmonics, can be used. Discuss the merits and demerits of the two options, and, from the discussion, explain which one is appropriate for high dynamic performance.
22. How can fundamental components be extracted from the switched machine input voltages? Will this affect the dynamic performance of the drive system?
23. The flux component of stator-current computation usually neglects the derivative term because it can produce a large flux current command in the indirect vector con-

troller for variations in the flux-linkages command. Is it safe to neglect it for small variations in the flux-linkages command?

24. Indirect vector controller is dependent on three motor parameters. Direct online measurement of these parameters is not possible. Indirect methods to detect all the parameter variations with one variable, such as reactive or real power errors, are easier. These variables are then used to adjust the slip-speed reference, for example. Justify with reasons how this can be adequate in parameter compensation, although individual parameter update is required in the indirect vector controller.
25. Enumerate all the possible signals that can be measured or extracted for use in the parameter adaptation of the indirect vector controller.
26. By adapting parameters in the vector controller, tuning is automatic, but compensating the parameter variations with only a slip-speed signal is not exact; it leaves out other input variables, such as i_f and i_T. Comment on the quality of the tuning of the vector controller achieved in this case.
27. Air gap power feedback control for parameter adaptation is better than modified reactive-power-feedback control. Justify it with reasons.
28. Will switching strategies affect the performance of the vector-controlled induction motor drive, particularly from the point of view of torque, speed, and position responses?
29. Develop a procedure to synthesize the current controllers, assuming that each phase current is individually controlled in the indirect vector-controlled induction motor drive.
30. Develop a procedure to synthesize the torque- and flux-component current controllers for a direct vector-controlled induction motor drive. What are the salient differences between these and the controllers discussed in question 29?
31. A flux-weakening controller is least parameter-sensitive in the stator and rotor flux-linkages-based vector controllers. Comment on the veracity of this statement.

8.18 EXERCISE PROBLEMS

1. Consider an induction motor whose parameters are given in worked Example 5.1 with $J = 0.01667$ kg-m^2 and $B = 0.0$ N·m/rad/sec. It is being driven by a current-source PWM inverter. It has a flux and torque processor for estimating the torque, the rotor flux linkage, and its angle. Assume that the control strategy used is the direct vector contrbl (per Figure 8.2) and the drive is in the torque mode with no dynamometer attached.
   (i) Develop a subsystem of the inverter with three inner current feedback loops. The inverter is operating in the PWM mode with a switching frequency of 6 kHz. The current error is amplified by a PI current controller (or, for ease, use a high-gain P controller) and limited to the rated current. Then this control signal is compared with the triangular PWM carrier signal to find the switching signals from which the applied line-to-line and hence phase voltages are found. Assume that the system is balanced.
   (ii) Use the algorithm given in the text and the block diagram shown in Figure 8.3 to realize the flux and torque processor.
   (iii) Once these subsystems are developed, close the torque and the rotor flux-linkages feedback loops to control torque and rotor flux-linkages. The flux-linkages

command is maintained at 0.405 Wb–turn and the torque command is set at 20.178 N·m. The PI (or P) limiters are set at 7.49 and 17.27 A for the flux- and torque-producing-component command currents, respectively. Then this gives rise to a peak stator-phase current of 18.84 A. Note that these are not rms values but peak values.

(iv) Test each subsystem before integrating the system completely. Plot the torque and its command, rotor flux linkage and its command, field and torque components of the stator current and their commands, stator-current phasor magnitude and torque angle, field angle, phase a current and its command, phase a voltage and rotor speed. Exercise care when you find the field angle by properly implementing the arctangent function for all quadrants.

2. Consider an indirect vector-controlled induction motor drive with rotor-position feedback. The inverter has three inner current feedback loops and works with a PWM frequency of 20 kHz. The parameters of the motor and drive are from the above problem. The field and torque current commands are limited to their rated and twice their rated values, respectively. The dc-link voltage can be considered to be 285 V, and the parameters of the controller match those of the machine. The load torque is zero throughout this problem. The following dynamic simulation results are required to enhance your understanding of the indirect vector-controlled induction motor drive:

   (i) With torque command kept zero, apply a rated rotor flux-linkages command. (The drive is in torque mode with speed loop opened.)

   (ii) When the steady state is reached in (i), apply a step torque command of 1 p.u. for 20 ms, then a step of -1 p.u. for 10 ms, and then 0 p.u. for 5 ms. (The drive is in torque mode with open speed loop.)

   (iii) Close the speed feedback loop with a high proportional gain in the speed controller and with the rated rotor flux linkages in steady state in the machine; then apply a step speed command of 0.75 p.u. for 100 ms, then apply a -0.75 p.u. for 50 ms, and then 0 p.u. for 10 ms.

   Plot all the key variables, such as the speed, torque and rotor flux-linkages commands and their responses, flux and torque current commands and their responses, phase a voltage and phase a current and its command. Plot these results in p.u.

3. Realize the indirect vector controller with discrete analog and digital integrated chips. Give only a block-diagram schematic of the realization. Minimize the number of components in the realization.

4. Stability is taken for granted, in general, in vector-controlled induction motor drives. Instability can arise when the controller and machine parameters do not match, as is shown in the case of the direct self control. Examine the stability of the indirect vector-controlled induction motor drive.

5. A perfect matching of machine and controller parameters is an ideal case that hardly ever arises in practice. Explore control strategies that use modern control theory to make the vector controller highly insensitive to machine-parameter variations.

6. Most of the parameter-compensation schemes are themselves parameter-dependent. Consider an example, and illustrate the effects on the performance of the indirect and direct vector-controlled motor drives.

7. Devise a control scheme to maximize the efficiency of the motor drive operating with a vector-control strategy.

CHAPTER 9

Permanent-Magnet Synchronous and Brushless DC Motor Drives

9.1 INTRODUCTION

The availability of modern permanent magnets (PM) with considerable energy density led to the development of dc machines with PM field excitation in the 1950s. Introduction of PM to replace electromagnets, which have windings and require an external electric energy source, resulted in compact dc machines. The synchronous machine, with its conventional field excitation in the rotor, is replaced by the PM excitation; the slip rings and brush assembly are dispensed with. With the advent of switching power transistor and silicon-controlled-rectifier devices in later part of 1950s, the replacement of the mechanical commutator with an electronic commutator in the form of an inverter was achieved. These two developments contributed to the development of PM synchronous and brushless dc machines. The armature of the dc machine need not be on the rotor if the mechanical commutator is replaced by its electronic version. Therefore, the armature of the machine can be on the stator, enabling better cooling and allowing higher voltages to be achieved: significant clearance space is available for insulation in the stator. The excitation field that used to be on the stator is transferred to the rotor with the PM poles. These machines are nothing but 'an inside out dc machine' with the field and armature interchanged from the stator to rotor and rotor to stator, respectively.

This chapter contains the description of the PM synchronous and brushless dc machines, derivation of their respective dynamic models, principle of control, analysis of the drive system, various control strategies, design of speed controller, flux-weakening operation, and position-sensorless operation. Some converter topologies for low-cost operation are also included at the end of the chapter.